SoC Labs · University of Southampton

MegaSoC

FPGA Prototyping & Hardware Bring-Up

Consolidated Work Record — Revision 8

Field	Detail
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Document date	30 October 2026
Period covered by this revision	29 July 2026 – 30 October 2026. Sections 3–13 are Revision 5 (to 30 September) carried over with update notes; Sections 14–20 are Revision 6 (1–21 October); Sections 21–26 are Revision 7 (21–28 October); Sections 27–32 cover 28–30 October 2026. Section 31 supersedes Sections 11, 19 and 25 as the current status.
Design	MegaSoC — Arm Cortex-A53 SoC, SoC Labs
Platform	Synopsys HAPS-SX VU19P 1F (xcvu19p-fsva3824-2-e); Vivado 2024.1 on the Linux server (sole build machine since 12 October); bench PC haps-dev for the two UARTs
Repository	github.com/SoC-Labs/MegaSoC-Project (public; the work folder since 22 October 2026; submodules MegaSoC-Tech, AHB-QSPI, SoCTools-Flow, ASIC-Flow). The FPGA work area FPGA_Tasos/haps_top/ holds Arm IP and stays on the server, never pushed; the bring-up contribution lives upstream under fpga/targets/haps_sx/bringup/.
Tape-out milestone	1 October 2026, IMEC
Headline change since rev3	The single open blocker in rev3 — "CPU never asserts ARVALID, root cause unknown" — is RESOLVED. Root cause found and fixed. Reliability confirmed at 100% (164/164 boot attempts, two independent hardware captures) once an unrelated measurement bug was also found and fixed.
Headline new work opened by this revision	A design-wide hold/removal timing audit (prompted by an independent constraints review) found the ARVALID fix was masking ~157,000 further real, previously invisible timing violations. Two of three root causes were fixed by 17 September (Section 5). The third, the QSPI clock, was resolved between 17 and 30 September by reclocking the controller on the system clock (Section 9.6); setup is now met with margin and the design has one clock domain.

Headline of the 17–30 September period	A structured review of every project-authored source, the synthesis and implementation logs and the routed netlist found 57 issues (Section 9 and the bug register in Section 10). The largest remaining problem is hold: about 20,000 endpoints failing by small amounts from clock skew across a design placed over two SLRs at 6% utilisation (Section 9.10). The boot console has still never been observed on hardware (Section 4.3), and no bitstream produced after early September has been run on the board.
Headline of the 1–21 October period (this revision)	The FPGA bring-up is complete. The boot console was observed for the first time on 10 October (build 42) once a polarity bug in the FPGA SRAM wrapper was found with the whole-bus logger (Section 14). Seven of the eleven SoC Labs tests then passed from RAM loaded over UART (build 48, 15 October), from an in-FPGA flash emulator (build 49) and from the physical on-board flash (build 51, 20 October) after two bugs in the QSPI controller's 1-bit write path were found and fixed (Section 17). The last blocker, a stop at cache enable present since build 42, was root-caused by RTL simulation of the FPGA design to Arm's TSMC example L2 tag RAM integration (written for a 1 MB L2, against this core's 256 KB) and fixed in build 54; on 21 October hello_world booted from the physical flash with both CPU caches on, the stock boot sequence end to end (Section 18).
Findings relevant to the ASIC from this period	Two bugs in the shared QSPI controller RTL (N39, one extra clock on every 1-bit write; N27, bit order within nibbles): no flash that needs a 1-bit status write can be programmed from the SoC until they are taken upstream. Firmware: scatter-loading never runs (N35, with an uninitialised global in the flash boot path); SPI_WAIT_BUSY can hang after a short command (N38); no Winbond branch in SPI_STARTUP (N26, validated branch supplied). Test code: uart_tests cannot pass on hardware (N40). Sections 17 and 19, bug register rows 51–68.
Headline of the 21–28 October period (this revision)	The bring-up was contributed upstream and merged (MegaSoC-Project PR #7, MegaSoC-Tech PR #4, 22 October), and the work folder moved from the GitLab clone to GitHub. The two SoC Labs tests still failing on 21 October were then root-caused and fixed. dap_tests: the DAP-Lite's system APB port was given the full system address instead of the offset within the 32 KB debug window, so every CPU access to the CoreSight ROM table was forwarded to the Cortex-A53 debug APB and answered with an error — a two-line change in megasoc_cpu_ss.sv, found by RTL simulation of the FPGA design and verified on build 57 (MBT-58, Section 22). uart_tests: six defects in the test itself, three of them timing races that only show with code fetched from flash (MBT-59, Section 23). Build 57, 27 October: all ten buildable SoC Labs test images pass from the physical on-board flash with no SRAM preload. Six fix pull requests (#5–#10) are open on MegaSoC-Tech for review and the two new findings have Jira tickets (MBT-61, MBT-62).

Findings relevant to the ASIC from 21–28 October	RTL shared with the ASIC: the debug-window address bug (MBT-58) means no self-hosted debug software can reach the CoreSight ROM table as integrated, and DBGROMADDR pointed at an unmapped address; the ROM table itself (Arm's DapRomDefs.v) is unconfigured and shadows the A53's own ROM table — an integration decision for SoC Labs (Section 22.3). Integration: megasoc_tech still pins the AHB-QSPI submodule at the GitLab-era commit that carries the two 1-bit write bugs (MBT-55/48), fixed on the AHB-QSPI master; the pointer must be bumped before the ASIC netlist is taken (row 77). Test code: timer_tests INTSTATUS race (MBT-61), uart_tests (MBT-59), c_print_str 79-character limit (MBT-62). The SoC Labs VCS flow aborts at the flash jump for every image, so flash-booted tests cannot yet be verified upstream (row 78).
Headline of the 28–30 October period (this revision)	The on-board DDR4 SODIMM is now the SoC's DRAM (Jira MGSOC-34). Step 1, 28 October: the Xilinx DDR4 IP, configured as the vendor's proven HAPS-SX design, calibrates on the board as a standalone bitstream and its traffic generator runs error-free. Step 2, build 58 the same day: the IP replaced the 512 KB block RAM on the NIC-400 DRAM port behind an asynchronous clock converter and a 64-to-512-bit width converter, and the Cortex-A53 loads and executes code from the SODIMM. The bench runs of 29 October then exposed a second instance of the N25 defect: the Vivado fileset still synthesised Arm's TSMC 1 MB example wrapper for the L2 DATA RAMs (build 54 had corrected the TAG RAMs only), so a dirty line evicted from the L2 was written back with another line's data — invisible until now because every RAM-mode run since build 47 had been executing with the CPU caches off. Build 59 (Arm's generic, index-masked wrappers for both RAM groups) passes the DDR4 test suite from DDR4 and from flash with the caches on, the seven SoC Labs RAM-mode tests execute from the SODIMM, and build 57's ten flash images still pass. Build 59 is the board baseline. Pull requests #5–#10 on MegaSoC-Tech were merged on 28 October; the DDR4 work and this record are contributed as MegaSoC-Project PR #8.
Findings relevant to the ASIC from 28–30 October	Nothing in this period changes the ASIC netlist, but three items need a check or a decision. (1) L2 RAM integration: the FPGA defect (row 79) is an integration file that selects the wrong-sized Arm example; the ASIC's own L2 data and tag RAM integration (mem_integrations/CortexA53 over the TSMC16 macros) must be confirmed to mask or size the set index for the 256 KB L2, since the same example tree is its starting point. (2) The NIC-400 decodes 64 KB of on-chip RAM at 0x80_0000 (row 85); the ASIC SRAM size should be checked against that decode. (3) Firmware library (Arm-licensed files, proposals only): enable_mmu() limits EL3 to 4 GB of VA and a 32-bit PA, so no stock test can reach DRAM above 4 GB (row 82); enable_caches() never sets CPUECTLR.SMPEN (row 81); boot.s enables the MMU only, so the SoC Labs tests run with the caches off unless the ROM enabled them (row 80). Also: SoC Labs' FPGA L2 data RAM wrapper in mem_integrations/FPGA stores 64 KB for a 256 KB L2 (row 84, unused here); the monitor's Winbond boot path jumps without the flash cache (row 83).

1. About this revision

This document is Revision 8 of the MegaSoC FPGA bring-up work record. It contains Revision 7 (28 October 2026) in full as Sections 1–26 — itself containing Revision 6 as Sections 1–20, Revision 5 as Sections 1–13 and Revision 4 as Sections 1–8 — with update notes in italics where later work changed the position, and adds Sections 27–32 for the period 28–30 October 2026. Section 31 supersedes Sections 11, 19 and 25 as the current status, and Section 32.1 records the positions of Revision 7 that this revision corrects. It supersedes nothing in Revisions 1–3 (consolidated as "MegaSoC_Work_Record_rev3.docx", dated 29 July 2026) — that document remains the authoritative history of everything done before 29 July 2026, and Section 2 of this revision summarises it rather than repeating it in full. This revision picks up exactly where rev3 left off and covers everything done since. Section 10 is a consolidated bug register for the whole programme: one row per bug, giving the bug, its root cause and the fix where one is implemented, with no reference to build numbers or source-tree versions; Section 10.7 adds the rows found in October.

Conflict resolution rule, carried over from rev3: where sources disagree, the most recent hardware evidence wins, and the superseded position is recorded explicitly (Section 7) rather than silently dropped. This revision overturns one of rev3's own headline numbers — the measured ARVALID reliability rate — and the reason it changed is itself one of the more important findings of this period (Section 3.3).

1.1 Source documents consulted

Source	Date	Role in this revision
MegaSoC_Work_Record_rev3.docx	29 Jul 2026	Authoritative prior history. Summarised in Section 2; its one open item is closed in Section 3.
MegaSoC_FPGA_Handover_Report.docx	Jul 2026 (rev 2)	Superseded by rev3; consulted only to confirm rev3's consolidation was complete and accurate.
MegaSoC_FPGA_Complete_Summary(_2).docx	Jun–Jul 2026	Superseded by rev3 (Q-channel root cause, later disproven). Consulted for extra technical detail (exact ILA capture values) not carried into rev3's condensed table.
MegaSoC_FPGA_BringUp_Complete.docx	Jun 2026	Earliest full log. Superseded on clock frequency, pin assignments and ILA status. Consulted for the full list of 12 SRAM macro names.
MegaSoC_FPGA_Work_Record.docx	27 Jul 2026	Programme/Jira context (task breakdown, owners, licensing, X2Go environment issues) not restated in as much detail in rev3. Summarised in Section 2.1.
PLL_Integration_Summary.docx	Jun 2026	DDR PHY PLL investigation, explicitly out of the author's main scope in every prior report. Summarised briefly in Section 2.6 for completeness only.
Interactive bring-up work, Linux server	7–17 Sep 2026	Sections 3–8 (carried over from rev4).
MegaSoC_HAPS_Bug_Register_2026-09-29.docx; external review note of 23 Sep	23–29 Sep 2026	Structured review of the project on the Windows workstation; 57 items, folded into Sections 9 and 10. The 23 Sep note drove the Linux-side changes and is superseded by the register.

Interactive bring-up work, both machines	17–30 Sep 2026	Sections 9–13.
MBT Jira board (soclabs.atlassian.net, project MBT)	29 Sep 2026	Tickets MBT-23 to MBT-36; cross-referenced in Section 10.
Interactive bring-up work, Linux server and bench PC haps-dev	1–21 Oct 2026	Sections 14–20. Per-build notes are archived with each bitstream under FPGA_Tasos/haps_top/bitstreams/build43_… to build54_….
MegaSoC_MBT_Jira_Updates _2026-09-30.md (running log, Desktop)	9–21 Oct 2026	Status proposals for MBT-23…36 and new tickets N24–N40 with evidence; Atlassian was unreachable from the server, so the text is paste-ready. Folded into Sections 10.7 and 19.
RTL simulations of the FPGA design (VCS) and unit testbenches (xsim)	14–21 Oct 2026	Sections 15.4 (QSPI controller), 16.1 (RAM slave), 17.1 (flash emulator), 18.2–18.3 (cache stop).
Interactive bring-up work, Linux server and bench PC haps-dev	21–28 Oct 2026	Sections 21–26. Per-build notes archived under bitstreams/build55_regression_20261022, build56_dap_fix_20261026 and build57_monitor_dapfix_20261026.
RTL simulations of the FPGA design (VCS), dap and uart testbenches	25–27 Oct 2026	Sections 22.2 and 23.3. The root cause of dap_tests and three of the six uart_tests defects came from them.
GitHub: SoC-Labs/MegaSoC-Project PR #7; SoC-Labs/MegaSoC-Tech PRs #4–#10	21–28 Oct 2026	Sections 21 and 24. Each PR description carries the verification status of its fix.
MegaSoC_MBT_Jira_Updates _2026-09-30.md (running log, Desktop); MBT Jira board	21–28 Oct 2026	Jira posts and ticket states (tickets to MBT-62); folded into Sections 10.7 and 24.
Interactive bring-up work, Linux server and bench PC haps-dev	28–30 Oct 2026	Sections 27–32. Per-build notes under bitstreams/ddr4_example_20261028, build58_ddr4_20261028 and build59_l2data_fix_20261029; ddr4/README_ddr4_step1.md and README_ddr4_step2.md.
Vivado hardware manager over JTAG from srv03335 (hw_mig calibration status, VIO probes)	28–29 Oct 2026	Sections 27.3, 28.4 and 29.4: calibration and margin read-outs of the DDR4 IP inside each bitstream.
xsim unit testbench of the DRAM bridge (ddr4/sim/tb_bridge.sv)	28 Oct 2026	Section 28.2: 203 checks of the clock/width converter chain against a behavioural model of the DDR4 IP.
Bench logs on haps-dev: ~/haps_captures/suite_20261029_* and suite_20261030_*	29–30 Oct 2026	Sections 29 and 30: every ddr4_fpga_tests run (versions 1–8), the RAM-mode regression from DDR4 and the flash regression on build 59.

MegaSoC_MBT_Jira_Updates _2026-09-30.md (running log, Desktop); MGSOC and MBT Jira boards; GitHub pull-request states	28–30 Oct 2026	Section 30.4 (PRs #5–#10 merged on 28 October) and the proposals of Section 32.3. Nothing was posted to Jira in this period; the log holds the paste-ready text.

2. Summary of prior work (Revisions 1–3, December 2025 – 29 July 2026)

Full detail is in MegaSoC_Work_Record_rev3.docx. This section is a condensed recap, included so this document stands on its own.

2.1 Programme and environment

MegaSoC is a three-year SoC Labs programme to design and fabricate a large, Linux-capable Arm-based SoC. The first milestone is a reduced-functionality tape-out to IMEC, deadline 1 October 2026 (Ethernet, PCIe and the chiplet interface are out of scope for this tape-out). The FPGA prototype's job is to prove the reduced SoC boots and behaves correctly before that database is committed. All work is tracked under Jira project MGSOC, parent task MGSOC-18. A dual-connection remote workflow (VS Code Remote-SSH, MobaXterm for X11, later an X2Go/XFCE desktop with its own diagnosed faults) was established against the development server srv03335.soton.ac.uk. Vivado ML Enterprise Edition licensing (the XCVU19P requires it; the free Standard edition does not cover this device) and a four-part VCS/Verdi simulation environment were both set up and documented as prerequisites.

2.2 Making the ASIC RTL synthesisable

- Twelve foundry SRAM macro behavioural models written (11 RFHSD_PG_SVT_* 1-cycle, 1 SRAMHDE_PG_SVT_* 2-cycle) — the A53 instantiates these by name and synthesis fails outright without them.
- Two cache/TLB RAM width parameter fixes: CPU_CACHE_PROTECTION=1 (80→84 bits) and CA53_SCU_L2_TAGRAM_DATA_W=40.
- Clock-gate cell PREICG_X4B_A9TS_C31.sv modelled from scratch (this file is revisited heavily in Section 4 of this revision).
- Clock controller bypass: FPGA_BUILD ties all system clocks directly to the single board clock for a synchronous bring-up domain; without this Vivado pruned most of the SoC as constant-driven logic.
- A 44-page platform reference document was produced for the HAPS-SX VU19P 1F board, because no consolidated textual pin map existed anywhere in the vendor documentation set.

2.3 Board bring-up and the elimination sequence

Pin assignments were corrected in three places (UART moved from unrouted/LED-conflicting pins to PMOD3); ILA debug infrastructure was built up to 16 probes; a ROM Q-channel deadlock was diagnosed and fixed (genuine bug, retained) but later shown, once observed against the fixed RTL, not to be the cause of the boot failure — the captured stall had been observed on a bitstream that predated the fix. The design's actual clock frequency was corrected from an assumed 20 MHz to the

real ConfPro setting of 100 MHz, then deliberately reverted back toward a lower frequency later for reasons specific to this SoC's own clock-gating structure and JTAG margin (see Section 4.3 of this revision, which documents the current, different, considered choice made in the XDC actually in use for the builds in this revision). By the end of rev3, every mechanism between power-on and the CPU's first bus transaction had been individually verified working with direct hardware evidence — clock, reset, ROM Q-channel, CA53 execution clock toggling, interconnect decode, boot ROM content — except one: the CPU's own AXI master port never asserted ARVALID. That single open question, MGSOC-33, is what Section 3 resolves.

2.4 Bugs found and fixed before this revision

#	Bug	File	Consequence if unfixed
1	ROM Q-channel ports tied off	megasoc_tech_wrapper.sv	PPU deadlock, ROM domain never sequenced on (real bug, later shown not to be the ARVALID cause)
2	SRAM external-gate override wired to the live power request	SRAM_wrapper.sv	Low-power sequencer could park, blocking all SRAM AXI traffic
3	Console output routed to a debug-only peripheral (USRT0 not UART0)	uart_stdout.c	No UART output would ever appear on the board pins
4	XDC clock period wrong relative to ConfPro's actual setting	haps_top.xdc	Design analysed at the wrong frequency
5	Clock-gate cells synthesising as LUTs on the clock tree	PREICG_X4B_A9TS_C31.sv	141 DRC critical warnings; ~20,000 failing setup paths at the time
6	Boot ROM image stale relative to a firmware fix	bootrom.v	A UART fix would never have been compiled into what the FPGA ran
7	Bootrom writes UART before configuring the GPIO pin mux	firmware	Latent silicon bug — early output lost on the real chip

2.5 Status at the end of rev3 (29 July 2026)

Verbatim from rev3's own headline:

"Design builds, programs and runs. CPU is powered, clocked and out of reset but never issues its first instruction fetch. Root cause open."

Every other subsystem checked was ruled out with direct hardware evidence: clock path, reset path, ROM Q-channel handshake, CA53 execution clock (confirmed toggling via a properly synchronised 2-flop toggle detector), boot ROM content (hand-decoded as valid AArch64), interconnect decode, and a full simulation boot through the SRAM memory test. The one untested lead going into this revision's period was the CPU_L2/CPU_NEON power/reset domains, sequenced by their own independent PCK-600 PPU instances.

2.6 DDR PHY PLL (excluded scope, noted for completeness only)

A separate task (dev_lpddr4_integration branch) to use the Synopsys DDR PHY's dwc_ddrphymaster_top as a behavioural clock source. The PLL's CSR registers are normally programmed by the PHY's own embedded PMU firmware (loaded from ICCM); tying CSR ports to constants does not configure it, so the PLL never locks that way. The delivered, working solution is a hybrid megasoc_clock_ctrl.v: self-oscillating clocks as the working source, with the real PLL instantiated and monitored in parallel and a USE_PLL_CLOCKS parameter to switch over once PllLockStatus asserts. This work is explicitly out of scope for the rest of this document, per the author's standing instruction in every prior report.

3. The ARVALID root cause — found, fixed, and confirmed at 100%

This section resolves MGSOC-33 and the single open item rev3 ended on. It is the headline result of this revision.

3.1 Root cause

The CPU's execution clock enable flop (ca53governor_cpu_clk_reset.v: stop_clk_reg) has an async PRE (preset) pin fed by a resync flop (u_ca53_cell_sync0/r1_reg) elsewhere in the same governor module. The timing relationship between these two flops includes a Removal check — the async-control-pin analogue of a hold check — which needs the data path delay between them to be large enough, not small (the opposite intuition to an ordinary hold/setup problem).

The XDC contained a blanket waiver, present from early bring-up and described in every prior report as “standard for large SoC prototypes on single clock”:

```
set_false_path -hold -from [get_clocks HAPS_CLK]
set_false_path -hold -to [get_clocks HAPS_CLK]
```

In Vivado, a -hold-class exception covers both ordinary Hold and Removal analysis. Because HAPS_CLK is the only declared clock in the design, this pair of lines silently excluded Removal analysis from this specific path — and, as later became clear (Section 5), from essentially every hold-type check in the whole design. No build had ever actually verified this path was safe; different builds' placement just happened to land on different sides of the real, unconstrained margin, which is why an earlier build (build 6, prior revision) had appeared to work and a later, nominally identical build (build 13) did not.

3.2 The fix: a placement pblock

Because the exception-precedence problem could not be solved by rewriting the timing exception itself (set_max_delay/set_min_delay were confirmed, empirically, to never override an existing set_false_path on the same path in this Vivado version, regardless of specificity), the fix actually used is a placement constraint instead — it cannot affect any other path's hold-waiver coverage, since it only tells the placer where two specific cells may go:

```
create_pblock pblock_governor_clkgate_proximity
resize_pblock pblock_governor_clkgate_proximity -add
{SLICE_X170Y440:SLICE_X230Y500}
add_cells_to_pblock pblock_governor_clkgate_proximity [get_cells -hier -filter { \
  NAME =~ "*u_governor_cpu_clk_reset/u_ca53_cell_sync0/r1_reg" \
  || NAME =~ "*u_governor_cpu_clk_reset/stop_clk_reg" \
}]
```

Several pblock sizes and an explicit LOC-pinned placement at the same measured distance were tried (20×20, 200×200, and a hand-picked site pair on the pblock's own best-measured row). None beat the original 60×60 region's own free-placement result; all were reverted. The lesson recorded from that experiment: Removal margin did not correlate monotonically with physical cell distance (+0.085 ns at 54 columns apart, −0.034 ns at 76 columns, −0.131 ns at 1 column) — clock-tree skew, not raw

distance, turned out to be the dominant term (this is explained structurally in Section 5.3). The 60×60 pblock, exactly as first written, is the configuration that is proven and retained.

3.3 The measurement bug that was hiding the true reliability rate

With the pblock in place, repeated hardware boot-cycle captures over several builds appeared to show roughly 78% reliability (28/36 pooled attempts asserting ARVALID/RVALID), which is what earlier iterations of this revision's own investigation initially reported and acted on — including a written summary sent externally describing a residual ~22% failure rate and several follow-up hypotheses (a second, structurally similar hazard in ca53_cpu_reset.v; the SCU tag-invalidate handshake). Both of those leads were investigated with real hardware correlation testing and found weak or inconclusive.

The actual explanation was found by accident while investigating those leads: the debug-UART capture process being used to gather this data was corrupted. Multiple orphaned cat /dev/ttyUSB0 processes, left running from earlier sessions across several days, were found still active and reading the same serial device concurrently, splitting the byte stream unpredictably between competing readers. This produced exactly the kind of garbled, dropped-byte log lines that had been quietly present throughout the whole investigation and were assumed, until this point, to be ordinary UART overrun.

Once every orphaned reader was killed and a single, clean capture was taken, the same 60×60-pblock configuration measured 51/51 boot attempts successful, then 113/113 on an independent repeat —

164/164 (100%) across two independent hardware captures.

The residual ~22% failure rate this investigation had spent real effort chasing (a second ca53_cpu_reset.v hazard, the SCU tag-invalidate handshake, various placement-tuning attempts) was never a real hardware defect. It was a measurement artefact of a broken capture process. This was independently reconfirmed on every subsequent build in this revision (builds 23 through 28 inclusive) that carried the same pblock: all measured 100%.

3.4 What this settles, and what it does not

- MGSOC-33 (“A53 boots correctly”), the gating hardware item in every prior report's checklist, is now supported by direct, repeated, clean hardware evidence: ARVALID and RVALID assert reliably, every boot, with the pblock fix in place.
- This says nothing about whether the CPU then executes correctly once it starts fetching — that is a separate question, addressed in Section 8.
- It also says nothing about the rest of the design's timing closure, which turned out to have its own, much larger, entirely separate problem — the subject of Sections 4 through 6.

4. Boot ROM content — verified and refreshed

A separate, shorter investigation, triggered by the natural next question once ARVALID was confirmed reliable: does the CPU then execute anything meaningful?

4.1 The FPGA "ROM" is a synthesised case-statement, not blank memory

ROM_wrapper.v (the copy actually compiled into this project, which has locally diverged from the shared megasoc_tech source tree) instantiates a bootrom module — an auto-generated Verilog case statement (bootrom_gen.py), not a $readmemh image, confirming the earlier finding that the ROM's contents are compiled directly into logic rather than loaded at runtime. The currently-embedded bootrom (dated 30 July 2026) was checked instruction-by-instruction: its first words decode as a standard AArch64 register-clearing startup sequence (0xd2800000 = MOVZ X0,#0, followed by a long MOV Xn,X0 run — a textbook C-runtime crt0 preamble), and the literal string "SoCLabs" is embedded in its data, confirming it is a real compiled image, not a placeholder.

4.2 Regenerated against the current bootloader build

bootrom_gen.py was re-run against the current bootloader.hex/SEC_ROM_BOOT_IMG. The result differed from the embedded bootrom by exactly one instruction word (plus the generation timestamp) out of roughly 8,200 — the embedded image was already almost exactly current. The freshly regenerated version was substituted in anyway, to guarantee the FPGA is running the definitively current software build rather than a near-identical one. This is build 24; it produced a clean, comparable timing result to build 23 (WNS=0.000 ns, WHS=+0.003 ns) and was independently reconfirmed at 100% ARVALID reliability.

4.3 Real boot-software console output — checked, not yet observed

Update, rev6: superseded. The console was first observed on 10 October 2026 (build 42) after the SRAM wrapper polarity fix, Section 14.

PMOD2 carries the SoC's real, software-driven UART0 output (distinct from PMOD3 IO[3], the custom debug_uart_probe.sv logger used throughout this whole investigation for hardware correlation captures). Whether real firmware output appears there was checked but not conclusively resolved: a genuine bootloader source exists (megasoc_tech/software/src/bootloader/bootloader.c, which prints a SoCLabs banner) and its content is confirmed embedded in the ROM, but no successful capture of visible output on PMOD2 was recorded in this revision. Given the bootloader's own object files touch the GIC-400, page tables and QSPI flash before reaching any UART print, a hang partway through that sequence is a plausible, untested explanation. This remains open — see Section 8.

5. An independent constraints review, and what it led to

After the ARVALID result was reached, an independent SoC-Labs constraints review of megasoc_haps_sx.xdc was obtained (artifact: “MegaSoC HAPS Constraints Review”). Several of its conclusions were cross-checked directly against this project's own RTL and real hardware/tool behaviour rather than accepted at face value; the review's central claims held up, with one number (an early estimate of ~140 unbuffered clock gates) self-corrected by the review itself once measured directly against a placed checkpoint, and the review acted on that self-correction.

5.1 Headline finding: the same waiver hid a design-wide problem

The review reached, independently and from SDC semantics rather than the empirical route this project had used, exactly the same conclusion as Section 3.1: -hold-class exceptions cover Removal as well as Hold, and the blanket waiver therefore hid every hold/removal check in the design, not only the one path already fixed by the pblock.

5.2 Removing the waiver: what was actually hiding underneath

The blanket waiver was removed (keeping the pblock exactly as-is, as an isolated experiment) and the design rebuilt with real hold/removal analysis for the first time. Result:

Metric	With blanket waiver (all prior builds)	Waiver removed, nothing else changed (build 26)
Failing hold/removal endpoints	0 (hidden, not actually zero)	157,405 – 168,745 (of ~244,000 total)
Total hold violation (THS)	0 (hidden)	−163,734 ns to −186,770 ns
Worst single path (WHS)	0.000 ns (hidden)	−3.719 ns

Vivado's own standard automatic hold-fix pass (phys_opt_design -hold_fix — never once run on this design in its whole history, because hold was always waived) was tried directly against this result and barely moved it (168,745 → 168,706 failing endpoints). This ruled out “ordinary local routing slop” as the explanation and confirmed the problem was structural.

5.3 Root cause 1 of 2: missing ASYNC_REG on the CA53's own synchroniser cell

The single worst path in the whole 168,745-endpoint picture was, itself, informative: not stop_clk_reg, but a different instance of the same underlying RTL cell — ca53_cell_sync.v, a plain 2-flop r0/r1 chain used 36 times across the CPU for exactly this kind of async-to-sync boundary, with no ASYNC_REG attribute anywhere in the RTL or the XDC despite genuinely being a synchroniser (r0's D input changes at the instant its own async clear releases — a textbook metastability-capture flop). A 3,000-path sample of the worst failing endpoints showed 94% landing in one module (u_ca53ifu_pf_bpd, the instruction fetch unit's branch-predictor table) — not because that module has its own defect, but because it happens to be enormous (6,488 flip-flops), so one bad net's fanout lands on thousands of individual destinations that all sit inside it.

Fix (build 27), applied via XDC rather than editing the licensed Arm RTL:

```
set_property ASYNC_REG TRUE [get_cells -hier -filter {NAME =~
"*ca53_cell_sync*/r0_reg" || NAME =~ "*ca53_cell_sync*/r1_reg"}]
```

Metric	Before (build 26)	After ASYNC_REG fix (build 27)	Change
Failing hold/removal endpoints	168,745	41,444	−75%
Total hold violation (THS)	−186,770 ns	−17,026 ns	−91%

5.4 Root cause 2 of 2: clock gates synthesising as LUT-driven global buffers

After the ASYNC_REG fix, the new worst path was of a completely different, ordinary kind (same-clock Hold, not Removal), between two flops with 5.449 ns of pure clock skew between them and no synchroniser involved at all — the signature the constraints review had predicted for its second claim: roughly 20 of this design's 24 global clock buffers (BUFGCE) are fed by combinational LUT logic rather than a clean source. This was fact-checked directly against a routed checkpoint using the review's own Tcl query and confirmed exactly: 20 LUT2-sourced, 3 clean (IBUFCTRL), 1 clean (BSCANE2 — the JTAG TCK, correct by design).

Root of the problem: PREICG_X4B_A9TS_C31.sv, the FPGA implementation of Arm's real ASIC clock-gate cell (instantiated ~100–140 times across the CPU/GIC/SCU), had already been through two prior revisions documented in rev3 — a behavioural latch+AND model (synthesises as a LUT directly in the clock path) and a blanket conversion to real BUFGCE (which exhausted clock-routing tracks and failed placement) — settling on a USE_BUFGCE parameter promoting only 2 of ~140 instances, leaving the rest on the LUT-based fallback. That fallback is exactly what the hold/removal audit was now measuring as a real, previously invisible defect.

A synthesis-level attempt was tried first: Vivado's built-in GATED_CLOCK_CONVERSION=auto option (a different, tool-native mechanism from either prior PREICG revision). It genuinely cleared a related but distinct DRC warning (PDRC-153, 0 occurrences afterward) by restructuring 143 combinational gated clocks onto flip-flop CE pins, but it left the specific LUT2→BUFGCE pattern completely unchanged (still 20 of 24) — the two mechanisms turned out to be independent, and this option was later found to be unstable across unrelated netlist changes (Section 6.4) and was ultimately not relied on.

Fix actually adopted (build 28) — PREICG_X4B_A9TS_C31.sv, revision 0.04, a third option nobody had tried: bypass gating entirely by default. Clock gating has no functional meaning on an FPGA (it is a power-saving feature, irrelevant until an ASIC pass needs it), so there is no correctness reason to keep any instance gated during bring-up, and a straight passthrough needs neither a LUT (no combinational logic left at all) nor a dedicated buffer (no new clock domain is created), sidestepping both of the two previous revisions' failure modes at once:

```
module PREICG_X4B_A9TS_C31 #(
  parameter USE_BUFGCE = 0,
  parameter FPGA_BYPASS_GATING = 1   // NEW, defaults on
```

```
) (
  output wire ECK, input wire E, input wire SE, input wire CK
);
  generate
    if (FPGA_BYPASS_GATING) begin : g_bypass
      assign ECK = CK;
    end else if (USE_BUFGCE) begin : g_bufgce
      BUFGCE u_bufgce (.I(CK), .CE(E | SE), .O(ECK));
    end else begin : g_behavioral
      /* original latch+AND model, preserved, now the fallback */
    end
  endgenerate
endmodule
```

Result (build 28): BUFGCE count across the whole device dropped from 24 to 9. Every single LUT2-driven instance in the CA53/governor/SCU family is gone — replaced either by a plain wire (no buffer needed at all) or, for the handful of genuinely high-fanout reset-distribution nets (the ASYNC_REG-tagged synchroniser outputs from Section 5.3), a clean FDCE-sourced buffer, which is the normal, correct pattern for wide reset fan-out.

Metric	Build 27 (ASYNC_REG only)	Build 28 (+ clock-gate bypass)	Change
Failing hold/removal endpoints	41,444	33,509	−19%
Total hold violation (THS)	−17,026 ns	−13,675 ns	−20%

Cumulative result, build 26 (waiver just removed) to build 28 (both fixes applied): 168,745 → 33,509 failing endpoints (

an 80% reduction

), and −186,770 ns → −13,675 ns total violation (

a 93% reduction

), from two targeted, well-diagnosed, independently-validated fixes — one a one-line XDC attribute, the other a single-file RTL parameter default. Build 28 is the

last known-good, fully rebuildable, stable baseline

 reached in this revision. It still shows 33,509 failing endpoints and is not fully timing-clean, but it builds reliably in normal time (no routing failures, no multi-hour anomalies) and its two fixes are both independently confirmed to be doing exactly what they were intended to do.

6. The QSPI clock-domain-crossing bug — real, partly fixed, not yet closed

With the two dominant causes fixed, the residual 33,509 failing endpoints were checked for concentration again (a 3,000-path sample): no single dominant cluster remained (the largest, QSPI, was 442 of 3,000 — 14.7%), spread thinly across dozens of modules instead. A second phys_opt_design -hold_fix attempt on this improved checkpoint again barely helped (33,509 → 33,363), ruling out ordinary routing slop a second time.

6.1 Root cause

The QSPI flash controller generates its own serial clock, QSPI_SCLK_i, by dividing HCLK with a runtime-writable register (qspi_clock_div.v, ratio set by software at any time):

```
assign QSPI_SCLK_i = (QSPI_CLK_DIV==5'h00) ? HCLK : QSPI_SCLK_reg;   // toggle-flop divider
```

This clock was never declared anywhere in the XDC. With no declaration, Vivado's static timing analysis does not recognise it as a separate clock at all — it traces electrical connectivity through the divider's mux and, for practical purposes, treats registers on QSPI_SCLK_i as if they shared HAPS_CLK's exact edges. This was confirmed directly: a register known from the RTL to be clocked by QSPI_SCLK_i (data_in_reg) showed up in Vivado's own timing report as “clocked by HAPS_CLK.” That mis-analysis is what produced the QSPI cluster of “violations” in the post-waiver-removal picture.

Independently of that constraint gap, the actual RTL crossings between HCLK and QSPI_SCLK_i were found to be genuinely unsafe: only a single flip-flop synchroniser in each direction (QSPI_enable_sync going in; AHB_QSPI_ENABLE_ACK and AHB_QSPI_BUSY consumed raw, with no synchroniser at all, coming back), where a correct crossing needs two stages. This is a real metastability risk independent of any timing-constraint question.

6.2 RTL fix applied — retained, considered correct, not the source of the later problems

Both crossings were widened to proper 2-stage (* ASYNC_REG = "true" *) synchronisers, in qspi_controller.sv (forward) and ahb_qspi_interface.sv (return), with the FSM logic updated to consume the synchronised signals. This RTL change alone (tested in isolation, Section 6.3, build 29) produces only a small, acceptable regression and is considered a genuine, retained improvement regardless of the outcome of Section 6.3–6.4.

6.3 XDC-side fix attempts — three tried, all unsuccessful, in a clear pattern

Getting Vivado to correctly treat the crossing as asynchronous (rather than leave it silently mis-analysed) proved much harder than expected, and produced the clearest and strangest finding of this revision.

Build	QSPI RTL sync (Section 6.2)	XDC treatment of the 3 sync paths	GATED_CLOCK_CONVERSION	Result
28	not present	—	auto	33,509 failing endpoints (stable baseline)
29	yes	create_generated_clock declared at a downstream leaf pin + set_clock_groups -asynchronous — did not propagate (only 1 of 1,434 registers on the net actually resolved to it); effectively no working exception	auto	38,509 (small regression — RTL change alone is nearly harmless)
30 / 30b	yes	set_false_path on each synchroniser's D input	auto	Build 30: routing never completed after 12+ hours, congestion spiking to 100,000+ overlapping nodes, killed. Build 30b: routing completed (~14.5 h) but left 8 nets with unresolved DRC RTSTAT-6 partial-route conflicts; bitgen refused to run.
31	yes	set_max_delay -datapath_only 40 ns on the same 3 paths (a softer exception, chosen specifically to avoid removing all timing pressure, after 30/30b's routing failures)	auto	Builds successfully this time, but 108,204 failing endpoints — a severe regression, over 3× worse than build 28. New LUT2-driven BUFGCEs appear in unrelated GIC-400 logic (FSM_onehot_state_reg[2]_i_2 ×5, priority_state_q_reg[7]_i_3) that were not present in any earlier build.
32	yes	same set_max_delay exceptions as build 31	off (to test whether the auto heuristic was the cause)	111,841 — still severely regressed, slightly worse than build 31. This disproves GATED_CLOCK_CONVERSION as the cause.

Reading across this table, the correlation is specific and repeatable: the RTL synchroniser hardening alone is nearly free (build 29). Adding any XDC timing exception at all on those same three paths — false_path or max_delay, and independent of the GATED_CLOCK_CONVERSION setting — triggers a severe, unrelated regression in logic (GIC-400 priority/state machine gates) that has no direct connection to QSPI. The mechanism for this is not yet understood; the most plausible explanation is that adding any exception at all on these paths perturbs Vivado's timing-driven synthesis/optimisation ordering enough to change unrelated automatic clock-buffer-insertion decisions elsewhere, but this has not been confirmed.

6.4 Current state — genuinely open

The next, not-yet-tried step identified and agreed at the point this revision was written: declare QSPI_SCLK_i as a generated clock correctly, at its true source (the qspi_clock_div toggle register's own output), rather than at a downstream leaf pin as build 29 mistakenly did — which should let it propagate network-wide the way it did successfully for HAPS_CLK itself, and may make a path-specific XDC exception unnecessary altogether. This has not yet been attempted.

Practical consequence for anyone picking this up: the live project source tree, as of this document, is in the state used for build 32 — i.e. it includes the QSPI RTL fix, and the regressed set_max_delay exceptions, with GATED_CLOCK_CONVERSION set to off.

This is not the last known-good state.

Build 28's XDC (waiver removed, ASYNC_REG fix, clock-gate bypass, no QSPI-specific exception at all) is the last configuration confirmed to build cleanly and reliably. Anyone continuing this work should either (a) revert to build 28's XDC as an immediate, safe baseline before attempting the QSPI fix again with the correctly-sourced generated clock, or (b) continue directly from the live tree with that understanding. This decision was still pending when this document was written.

Update, rev5 (30 September): this section is superseded. The QSPI clock was removed altogether rather than constrained: the controller now runs on the system clock with edge enables (Section 9.6). The set_max_delay lines discussed above were later found never to have been in force (Section 9.9).

7. Superseded findings — corrections to this revision's own earlier positions

Recorded explicitly, in the same spirit as rev3 Section 8, because intermediate conclusions reached during this revision's own period were themselves overturned by later evidence within the same period.

Earlier position (this revision)	When	Current position
Reliability with the pblock fix is ~78% (28/36); residual ~22% failures are a real, unexplained hardware defect worth chasing with new debug taps	Early in this revision, reported externally	The pblock fix alone measures 100% (164/164 across two clean captures). The ~22% was a debug-UART capture artefact (multiple orphaned concurrent readers), not a hardware defect. See Section 3.3.
The second ca53_cpu_reset.v hazard and the SCU tag-invalidate handshake are plausible contributors to the residual failure rate	Mid-revision	Both leads are moot — there was no residual failure rate to explain once the measurement bug was fixed. Neither was conclusively confirmed or refuted before this was discovered.
GATED_CLOCK_CONVERSION=auto is likely the cause of the severe regression seen when adding QSPI XDC exceptions	After build 31	Disproved directly: build 32, identical except GATED_CLOCK_CONVERSION=off, regressed just as badly. See Section 6.3/6.4.
A create_generated_clock declared at a downstream consumer pin should correctly describe QSPI_SCLK_i	Build 29	Confirmed not to work — it does not propagate to the rest of the net. Must be declared at the true source instead (Section 6.4, not yet attempted).

8. Project portability — moving the project to a second machine

The project was copied to a Desktop folder for transfer to a different PC (university machine), which surfaced and led to fixing real, pre-existing project-hygiene defects unrelated to anything above, worth recording because they will recur for the next person who copies this project.

8.1 What was copied

haps_top.xpr, haps_top.srcs (all sources, including every locally-materialised file edited in this revision), haps_top.cache, haps_top.ip_user_files and haps_top.gen — 557 MB in total. haps_top.runs (build artefacts/bitstream) was deliberately excluded: it is not needed to rebuild from source, and was mid-write by an in-progress build at the time of transfer.

8.2 Broken fileset entries found and fixed

Opening the transferred project on the second machine produced “could not find the file” errors for five specific files (Axi.v, Axi_undefs.v, AhbDefns.v, p_flash_cache_f0_gen_const_pkg.vh, ca53ifu_defs.v). Root cause: each of these files has a duplicate entry in haps_top.xpr, one portable ($PSRCDIR/...) and one pointing directly at the original Windows machine's path (C:/Users/Tasos/megasoc_tech/...) or the shared /research/AAA/ library — neither of which exists on a different PC. This class of duplication (multiple fileset entries for the same physical file) is pervasive in this project — 136 files in total carry an AutoDisabled="1" duplicate entry — but a systematic check confirmed only these 5 files had the specific, broken pattern of zero enabled copies remaining after the non-portable entries were removed; the other 131 are legitimately, harmlessly disabled (simulation-only testbenches, ASIC-only tsmc180 technology cells, unused alternate peripheral variants).

First fix attempt made this worse: deleting the non-portable duplicate for Axi.v/Axi_undefs.v left only their AutoDisabled="1" portable copy behind, meaning the file existed nowhere active in the synthesis fileset — which surfaced as a second, different error (“cannot open include file”, 40 instances, since 68 NIC-400-generated files include these two headers by bare filename). Corrected by removing AutoDisabled="1" from the surviving portable entries instead of deleting anything further. Verified directly against Vivado's own resolved project state (IS_ENABLED property), not just by re-reading the XML, after both the mistake and the correction.

9. Structured review and consolidation (17–30 September 2026)

Two things happened in this period. First, the project was ported to a second machine (Windows workstation, Vivado 2026.1) and put through a structured review: static reading of every project-authored and hand-forked source, the synthesis and implementation logs, queries against the routed netlist, and a full disassembly of the boot ROM from the reset vector to UART initialisation. That review produced the HAPS Bug Register of 29 September (57 items). Second, an external review note of 23 September proposed a set of fixes that were applied independently on the Linux server. The two lines of work were reconciled on 30 September (Section 9.12).

Unless a subsection says otherwise, every fix in this section is verified by synthesis and implementation reports and by static analysis only. None has been confirmed on hardware.

9.1 Clock gating: bypass extended to the power controllers and the GIC

Section 5.4 bypassed the CA53's own PREICG clock-gate cells. Two further families were still gated: pck600_clock_gate (the PCK-600 power controllers that sequence CPU power-up and reset release) and gic400_inter_clkgate (the GIC-400 distributor). Both kept the behavioural latch-and-AND model, which on FPGA becomes a latch and a LUT in the clock path. Measured on the routed design, 144 of the 200 worst hold paths ended in the power controllers clocked this way; after bypassing both cells, none did. Two defparam lines that tried to force USE_BUFGCE on the PREICG cells were found to be unreachable and were deleted.

9.2 Reset structure

Three resets were formed by combinational ANDs of power-controller outputs and fed straight to asynchronous reset pins: ROM_RESETn, SRAM_RESETn and the CA53's nCORERESET (the latter from the August experiment described in Section 3). A LUT ahead of an async reset can emit a runt pulse when its inputs change on the same edge, and the release edge was unsynchronised. The ROM and SRAM resets now assert asynchronously and release through a two-flop synchroniser on the peripheral clock. The CA53's four reset inputs are driven from a single signal, asserted asynchronously and released together after a 16-cycle hold, as the Cortex-A53 TRM requires. One side effect is recorded: a warm reset now also asserts the power-on and debug resets, so debug state is lost on a warm reset. Acceptable for bring-up; to be reviewed for the ASIC.

9.3 ROM gate request

A debug counter, left over from an earlier investigation, drove the ROM controller's external gate request low for the first 64 cycles after reset release and for 64 of every 512 cycles thereafter, ignoring the accept signal. The first window coincides with the CPU's first fetch and the EL3 page-table walk. The controller's gated response path is configured never to answer, so each window simply stalled ROM traffic. The request is now held at 1 and the counter is gone.

9.4 L2 data RAM model

Update, rev6: the two-cycle model introduced here was itself wrong (one cycle late against Arm's generic model) and was corrected in build 53; it was not the cause of the cache-enable stop, which was the L2 tag RAM integration (Section 18).

The two-cycle SRAM model used for the sixteen L2 data banks advanced its output stage only while the chip enable was active in the same cycle. A read followed by an idle cycle therefore never reached the output; the data stayed in the first stage until the next access. The output stage is now advanced by the previous cycle's enable. Checked in simulation with an isolated read.

9.5 Peripheral subsystem

Three wiring defects fixed: the watchdog's clock, clock-enable and reset inputs were unconnected (interrupt unknown in simulation, dead on FPGA); the AHB default slave sat on a disabled mux port, so every unmapped access in the peripheral window completed OKAY with data 0; and the default-slave select was not qualified by the bus select, which would have hung other targets once the port was enabled. Unmapped peripheral addresses now return a bus error. The expansion-window BVALID tie (a permanent unsolicited write response) was set to 0. The error slave on the DRAM port accepted write data before an address and then hung; it now waits for AWVALID.

9.6 QSPI: the clock removed rather than constrained

Section 6 left the QSPI clock open. Every attempt to describe it in constraints had either failed to propagate, made routing diverge, or (as found in this period) been rejected by the tool without anyone noticing. The clock itself was the problem: a runtime-selected mux between the system clock and a divided flop, clocking the controller on both edges, with 186 latches inferred from incomplete combinational blocks. Measured on the routed netlist, 901 register and latch pins had no clock at all, and after Section 5's fixes all 200 of the worst hold paths ended inside the controller.

The fix removes the clock. The divider now produces a registered SCLK level plus two single-cycle enables on the system clock, asserted on the edge where SCLK changes. Every register that was clocked on an SCLK edge is now clocked on the system clock and enabled by the matching pulse, so it updates at the same instant as before. The passthrough case (divider value 0, SCLK equal to the system clock) cannot be expressed as an enable and now behaves as divider value 1, i.e. half the system clock; this is a behaviour change firmware could notice. All 186 latches were converted to registers or fully specified logic. Result on the routed design: setup went from 0.000 ns with much of the block outside analysis to +25.9 ns on a 50 ns period with every path timed; the design's clock domains went from two to one; inferred latches from 186 to 0. The logic is not simulated: the project testbench cannot reach the flash path. Its first real test is hardware.

9.7 Debug logger

The UART event logger that produced every hardware measurement in Section 3 was itself found to have three defects. It drains one event per line (about 1.38 ms) with strict lowest-ID-first arbitration, and the sticky ARVALID flag sat below two per-transaction signals that are quiet only while the CPU is stuck. The logger therefore went blind exactly when the CPU began fetching, and a starved slot and a never-asserting ARVALID produced identical logs. Separately, the clock-activity flag false-positived after every reset because its synchroniser flops were not reset while its compare register was; and the edge detector's reference register reset to zero while the live bus did not, producing about 10 ms of fabricated events at every reset release. The ARVALID flag is now at ID 0, an overflow flag is reported on ID 1, the detector is armed one cycle after reset, and the synchroniser flops are reset. The 100% figure in Section 3.3 was measured after the capture process was fixed, and the starvation

defect could only have hidden a positive result, so the figure stands; it has not been re-measured on any later build.

9.8 RT clock

The real-time clock fed to the RTC and timers was a counter bit at 305 Hz, where 32.768 kHz is intended, and the one-hertz clock derived from it inside the peripheral subsystem ticked once per 107 seconds. Neither was declared as a clock, so 47 register pins, including the RTC's 32-bit seconds counter, sat outside timing analysis. The top-level divider is now a divide-by-610 toggle register declared as a generated clock. The one-hertz clock inside the peripheral subsystem is still undeclared. Confirmed by ROM disassembly not to affect boot.

9.9 Constraints that were silently not in force

Three cases were found where a constraint had done nothing without anyone knowing. The three set_max_delay -datapath_only lines of Section 6.3 had been rejected on every build for lacking -from (Constraints 18-540), so the experiment that compared them with set_false_path had in fact compared set_false_path with no constraint at all. A Tcl procedure written into the constraints file as a guard against exactly this class of failure was itself rejected by the project flow (Designutils 20-1307), and every constraint that used it was dropped; the guard now lives in the build script, which stops between synthesis and implementation if an expected clock is missing or the log contains a rejected-constraint warning. The CLOCK_DEDICATED_ROUTE property on the clock input buffer finds no object during synthesis and is probably obsolete.

No I/O port had timing constraints. Constraints for the flash pins were added, referenced to a generated clock at the SCLK pad (following D. Mapstone's constraints for the same IP on Pynq-Z2). The first version specified hold requirements the hardware does not need: the controller changes data on the SCLK falling edge and the flash samples on the rising edge, half a period later, but that relationship is guaranteed by the enable structure, not by anything static timing can see, so the tool demanded same-edge hold and reported 319 pins it could not fix. Hold is now excluded on the flash data pins, setup is kept, and chip select is a false path. These constraints are implementation-only, because the pad clock cannot be resolved during synthesis.

9.10 Placement and the remaining hold problem

With the structural sources of hold failure removed, what remains is broad and shallow: about 20,000 endpoints failing by small amounts across the NIC-400, the power controllers, the SD controller and the AXI bridges, which the router reports it cannot fix. The clock utilisation report explains it: the single 20 MHz clock drives about 89,000 loads over 57 to 59 clock regions, and a per-cell placement count shows the design occupying SLR0 and SLR1 with nothing in SLR2 or SLR3, while using 6% of the LUTs and 1% of the registers. The worst individual paths cross the SLR boundary. A router option suggested by the tool itself (global hold iteration) was tested and produced a bit-identical result; post-route physical optimisation with hold fixing recovered about 2 ns.

A floorplan constraint confining the SoC to two SLRs is prepared but not yet enabled. The first draft named SLR1:SLR2 on the assumption that the design spanned all four SLRs; measured directly on this part, clock-region rows Y0–4 are SLR0, Y5–9 SLR1, Y10–14 SLR2 and Y15–19 SLR3, each with 540 block RAM sites, and the design already sits in SLR0 and SLR1. The range was corrected to SLR0:SLR1.

Since the design is already there, a tighter region within those two SLRs is what would reduce the clock span. The 487 block RAMs used are 45% of two SLRs' capacity and 90% of one, which is why one SLR is not an option.

9.11 Methodology audit

A fresh report_methodology on a routed design of this period listed 700 items. Most are pre-existing and benign (216 byte-write-enable notes on the CA53 cache RAMs; 265 large-skew hold paths, all the same pattern as Section 9.10; nine LUT-driven async resets inside licensed Arm RTL). Three are worth recording. The RTC's 32-bit counter had no timing clock at all (32 registers, critical). Vivado's own CDC check flagged the QSPI crossing as unsynchronised, and flagged the pair-wide datapath-only exception that had been tried on the Linux server as a masking risk; both are moot now that the clock is gone. And the first version of the CA53 reset fix on the Linux server still left a two-input LUT driving the core reset pin; the single-signal version adopted in the merge does not.

9.12 Two source trees, and the merge

Between 17 and 29 September the Linux server and the Windows workstation each carried fixes the other lacked, and solved two problems (the CA53 reset, the logger) in different ways. On 30 September the trees were diffed file by file: 38 files differed. The Windows tree was taken as the base, because it carries the QSPI rewrite; the RT clock fix and the build-script guard were carried over from the Linux tree, and the Linux versions of the reset and logger were dropped. The Linux tree's sources were archived before the merge. Which machine builds from here is an open decision; the two should not diverge again.

9.13 Timing progression in this period

Stage	WNS (ns)	WHS (ns)	Latches	Critical warnings
Start of period (Section 5 fixes in place)	0.000	-4.592	9	4
+ power-controller and GIC gating bypassed; resets; ROM stall; L2 RAM; logger; peripherals	0.000	-3.524	7	4
+ QSPI reclocked; all latches removed	+25.909	-0.664	0	0
+ flash I/O constraints; post-route hold fixing	+18.008	-0.856	0	1
+ flash pad hold exclusion (Windows)	running at time of writing		0	
Merged tree + RT clock (Linux, Vivado 2024.1)	+13.309	-0.649	0	0

The 0.000 ns setup figures in the first two rows are not real margins: much of the QSPI block was outside analysis. The +25.9 ns figure in the third row is the first setup result with every path timed. In the merged build the setup-limiting path is the flash input (pad clock to system clock, 13.3 ns); internal setup margin is 21.7 ns. Its worst hold is -0.649 ns, with 44,955 endpoints failing by a total of 13,686 ns, and the router reports 19,665 pins it cannot fix - the same population as on the Windows builds. No LUT remains anywhere on a clock tree.

10. Bug register

One row per bug found in the programme to date, from the first board bring-up to 30 September 2026. Each row gives the bug, its root cause, and the fix where one is implemented. Rows with an empty Fix column are open. The register is organised by area, not by date, and deliberately carries no build numbers or source-tree versions; those are in Sections 3–9 and Appendix B.

10.1 CPU boot

#	Bug	Root cause	Fix
1	CPU never issued its first instruction fetch (ARVALID never asserted); one build worked and a nominally identical one did not.	The execution clock-enable flop in the CA53 governor has an async preset fed by a resynchroniser flop. The Removal check on that path was waived by the blanket hold waiver (row 2), so placement landed on either side of the real margin from build to build.	Placement constraint keeping the two flops together. Confirmed 164/164 boots. Now redundant: the clock-gate bypass (row 6) removed the flop, and the constraint should be deleted.
2	Blanket set_false_path -hold on the only clock.	Added early as "standard for single-clock prototypes". A hold-class exception also covers Removal, so every hold and removal check in the design was waived.	Removed.
3	CA53 two-flop synchronisers (36 instances) could be separated or retimed by the tools.	No ASYNC_REG attribute in the licensed RTL.	ASYNC_REG applied from the constraints file.
4	Boot reliability measured at 78% when it was 100%; weeks spent on two false leads.	Orphaned serial readers on the host split the UART byte stream between them.	Kill stale readers before capture. Clean captures measured 51/51 and 113/113.
5	Real boot console (UART0) never observed on hardware.	Unknown. The boot ROM touches no peripheral before UART init and the banner is due within about 350 us of reset, so silence means the CPU stops early.	

10.2 Clocking

#	Bug	Root cause	Fix
6	CA53 clock-gate cells synthesised as LUTs in the clock path (about 140 instances); the tool promoted some to global buffers fed by a LUT, a different set on each build.	The ASIC clock-gate cell was modelled as a behavioural latch and AND. Clock gating has no function on FPGA.	Gating bypassed (clock passes straight through). Must be reverted for the ASIC.
7	Power-controller (PCK-600) and GIC-400 clock-gate cells: same defect; 144 of the 200 worst hold paths ended in the power controllers.	Same latch-and-AND model, missed when the CA53 cells were converted.	Both bypassed. Must be reverted for the ASIC.
8	Two defparam overrides intended to promote clock gates to global buffers had no effect, and the constraints comments described a clock structure that was not built.	The generate condition they targeted was already taken by another parameter; defparam on a generate condition is not legal.	Deleted.
9	QSPI controller clocked by a runtime-selected mux between the system clock and a divided flop, on both edges; 901 register pins had no clock; every constraint attempt failed or was rejected.	A clock built from fabric logic cannot be described to static timing, and the passthrough arm made the divider flop's clock ambiguous.	Controller reclocked on the system clock with rise and fall enables; passthrough removed (divider value 0 now behaves as 1). Not simulated; first test is hardware.
10	186 latches inferred in the QSPI controller and its register block.	Incomplete always_comb blocks (read data, register readback, last address, FSM next state, acknowledge).	Converted to registers or fully specified logic; zero latches.
11	RT clock 107 times too slow (305 Hz for 32.768 kHz); one-hertz clock ticks every 107 s.	Placeholder divider took a counter bit.	Divide-by-610 toggle register, declared as a generated clock.
12	RT clock and the one-hertz clock unconstrained; the RTC's 32-bit counter had no timing clock at all.	Counter bits used as clocks with no create_generated_clock.	Top-level RT clock declared. One-hertz clock inside the peripheral subsystem still undeclared.
13	Design placed across two SLRs and 57 to 59 clock regions at 6% LUT utilisation; about 20,000 hold endpoints the router cannot fix.	Clock skew scales with clock-tree span; the worst paths cross the SLR boundary. No placement guidance was given.	Floorplan constraint prepared (SLR0:SLR1, corrected from a draft that named the wrong SLRs). Not yet enabled.
14	Simulation and FPGA use different clock topologies.	The behavioural file list drives three unrelated clocks; the FPGA ties CPU, system and flash clocks to one net.	Documented. Check which file list a simulation used before treating it as representative.
15	Constraints file analysed the design at the wrong frequency.	Period written before the board clock was settled.	Period corrected to the programmed clock.

16	ILA and debug hub still present in every build, a second (JTAG) clock domain, despite being recorded as removed.	Instance left in the ROM wrapper under an ifdef.	Removed. The routed design has one clock.

10.3 Reset

#	Bug	Root cause	Fix
17	ROM and SRAM block resets were a LUT AND of power-controller outputs driving asynchronous reset pins; release unsynchronised.	No reset synchroniser; two flops changing on the same edge in opposite directions can produce a runt pulse.	Asynchronous assert, two-flop synchronous release on the peripheral clock.
18	CA53 power-on reset released about 5 cycles before the L2 reset, contrary to the TRM.	Core and L2 power controllers are independent state machines with no ordering between them.	All four CA53 reset inputs driven from one signal: async assert, 16-cycle synchronous hold on release.
19	The interim core-reset workaround made core reset a permanent function of L2 reset, and formed it as a LUT on an async pin.	Combinational AND of two power-controller outputs.	Superseded by row 18.
20	Top-level auto-reset comparator output mixed into an async-reset branch without being in the sensitivity list; board reset used with no debounce and asynchronous release.	Bring-up convenience logic.	

10.4 Memory and Q-channel

#	Bug	Root cause	Fix
21	ROM Q-channel ports tied off; ROM power controller deadlocked; CPU stalled on its first fetch.	Tie-offs in place of the power-controller connection.	Q-channels connected to the ROM power controller.
22	SRAM low-power sequencer could park, blocking all SRAM traffic.	External gate override wired to the live power request instead of held inactive.	Held at 1.
23	ROM stalled for the first 64 cycles after reset (the first-fetch and page-table-walk window) and 12.5% of cycles thereafter.	A debug counter drove the external gate request, ignoring the accept signal; the gated response path is configured never to answer.	Request held at 1; counter removed.
24	Two-cycle L2 data RAM model returned stale data for any read not followed by another access.	Output stage advanced only while the enable was active in the same cycle.	Output stage advanced by the previous cycle's enable.

25	ROM and SRAM controller port widths: write enable 8 bits into a 9-bit port; SRAM data 64 bits into 65-bit ports. The parity lane is undriven while cache protection is enabled.	Wrapper signals not sized for the parity bit.	
26	Memory controller low-power state machine has an unused state that decodes to accept and deny at once, and a default branch assigning X.	Licensed IP; no recovery from an upset.	Not editable. Proposed: bound assertion that the state is never reached.
27	RAM shim outputs tied to 0 where 0 is the asserted level (unverified).	Placeholder tie-offs.	

10.5 Bus and peripherals

#	Bug	Root cause	Fix
28	Expansion-window write response tied permanently valid: an unsolicited response into the interconnect.	Constant tie-off on a stateful signal.	Tied to 0.
29	Any access to the expansion window is accepted and never answered (hangs).	Ready signals tied 1 and valid signals tied 0. The upstream fixes (outstanding-request counters, ID echo) were never ported to the FPGA fork.	
30	Unmapped peripheral addresses completed OKAY with data 0.	AHB default slave on a disabled mux port; default-slave select not qualified by the bus select.	Port enabled; select qualified. Unmapped addresses now return a bus error.
31	Watchdog interrupt unknown in simulation, dead on FPGA.	Watchdog clock, clock-enable and reset inputs left unconnected.	Connected.
32	Error slave on the DRAM port hung after a write whose data arrived before its address.	WREADY asserted unconditionally in idle, so the beat was consumed and the state machine waited for a second last beat.	WREADY waits for AWVALID.
33	DRAM port had nothing answering it.	DDR4 not integrated on the board; the DRAM wrapper was commented out.	Error slave returning DECERR added as a terminator.
34	Interrupt synchroniser into the GIC-400 is a single flop; RTC-domain interrupts cross asynchronously.	One stage on 240 lines. The GIC requires inputs synchronous to its clock.	Not yet ported: the two-stage version exists upstream.
35	Undriven and unconnected ports: SPI readback upper half; USRT divider output; CPU memory-attribute outputs.	Incomplete wrapper wiring.	

36	Console output routed to a debug-only peripheral (USRT0, not UART0); nothing could reach the board pins.	Wrong peripheral in the printf retarget.	Retargeted to UART0.
37	Boot ROM image lagged a firmware fix.	ROM not regenerated after the firmware changed.	Regenerated from the current firmware image.
38	Boot ROM writes to the UART before configuring the pin mux; early output lost on real silicon.	Initialisation order in firmware.	Not confirmed fixed (tracked separately as MGSOC-49).

10.6 Debug instrument and tooling

#	Bug	Root cause	Fix
39	Debug logger could not report ARVALID at the moment it mattered; a starved slot and a never-asserting signal produced identical logs.	One event per line with lowest-ID-first arbitration; the ARVALID flag sat below two per-transaction signals.	ARVALID flag moved to ID 0; overflow flag on ID 1.
40	About 10 ms of fabricated logger events at every reset release.	Edge-detector reference register reset to zero while the live bus did not.	Detector armed one cycle after reset.
41	Clock-activity flag went true on every boot after the first.	Synchroniser flops were not reset while the compare register was.	Flops reset. The flag is now always true under gating bypass and measures nothing; still to be repurposed.
42	Logger loses one edge permanently for a bit that pulses while queued; timestamps taken at FIFO push, not detection.	Per-bit event queue.	Not fixed in the current logger (a whole-bus format that removes the queue was prototyped and set aside).
43	Reset debug tap is a separate AND gate, not a tap on the reset net that reaches the CPU; it cannot fail.	Tap duplicated the logic instead of observing the net.	
44	JTAG/ILA readback corrupt at 20 MHz (frozen probes, glitches on unrelated signals).	Debug hub behaviour at this clock rate; not root-caused.	UART logger used as the only debug channel; ILA removed.
45	Testbench cannot fail: no $fatal or $error; a dead CPU and a clean boot both exit 0. Monitor watches an SRAM address; debug UART unconnected; simulation lacks the FPGA_BUILD define; no flash model.	Bench written for waveform inspection, not regression.	
46	Project would not open on a second machine (missing files, missing include).	Duplicate file entries, one with a machine-specific path and the portable copy disabled.	Duplicates removed; surviving entries enabled.

47	Under Vivado 2026.1: include path pointed at a file; 17 SD-card files set default_nettype none and never reset it, breaking the Arm memory-controller files compiled after them; an old incremental checkpoint was reused; a run property with a space and a value beginning with a dash aborted the script.	Environment differences between tool versions.	All four corrected.
48	Three constraints were never in force: set_max_delay -datapath_only without -from (rejected every build); a Tcl procedure in the constraints file (rejected, and every constraint using it dropped); CLOCK_DEDICATED_ROUTE finding no object.	Critical warnings not checked between synthesis and implementation.	First two removed or moved to the build script, which now stops on a rejected constraint. Third open.
49	Flash pad constraints demanded hold margin the hardware does not need; 319 pins the router could not fix.	Data changes on the SCLK falling edge and is sampled on the rising edge, a relationship guaranteed by the enable structure and invisible to static timing.	Hold excluded on the flash data pins; setup kept; chip select false-pathed.
50	Global hold iteration router option had no effect.	The router gives up on violator count before consulting it.	Ruled out; do not retry.

11. Current status — done and outstanding

Update, rev6: this section is the position on 30 September and is superseded by Section 19.

Task / action	Status	Note
CPU issues its first instruction fetch (MGSOC-33)	DONE	Section 3. 164/164 boots. Not re-measured since early September; the debug logger that measured it has since been corrected (Section 9.7).
Real boot-software console output (UART0)	OPEN	Never observed. The current boot question. Section 4.3 and bug 5.
Design-wide hold/removal audit	IN PROGRESS	Structural causes removed (Sections 5 and 9). Setup met with margin. About 20,000 shallow hold failures remain from clock skew (Section 9.10); floorplan prepared, not tried.
QSPI clock	DONE (unsimulated)	Controller reclocked on the system clock; zero latches; one clock domain (Section 9.6). Hardware test pending.
Reset structure	DONE (static only)	Section 9.2. Hardware test pending.
Debug logger trustworthy	DONE	Section 9.7. Conclusions recorded before 22 September against the old logger should be read with that in mind.
Single source tree	DONE	Merged 30 September (Section 9.12). Decide which machine builds from here.
Fork reconciled with upstream	OPEN	At least two upstream fixes missing (bugs 29, 34).
Testbench able to fail	OPEN	Bug 45.
Tape-out revert register	OPEN	Section 12. No named owner or gate.

12. Updated tape-out revert and review register

Rev3 Section 9's 15 items and rev4's items 16–23 (reproduced in the previous revision) still apply, with the changes below. New items from this period follow.

#	Item	Type	Action before tape-out
16 (changed)	Governor clock-gate proximity pblock	FPGA-only placement constraint	Now constrains one cell and does nothing (the other was optimised away). Remove.
18 (changed)	Clock-gate bypass, now in three cell families: PREICG, pck600_clock_gate, gic400_inter_clkgate	FPGA-only RTL change	All three must be reverted for the ASIC. The originals are kept in the files.
19–20 (changed)	QSPI synchronisers and constraints	RTL / XDC	Superseded: the controller is reclocked on the system clock. The reclocked controller is a functional change (divider value 0 behaves as 1) and must be reviewed for the ASIC, not assumed FPGA-only.
24	Reset synchronisers on ROM and SRAM block resets; single synchronously released CA53 reset	RTL	Retain for the ASIC after review. Note the warm-reset side effect (Section 9.2).
25	ROM external gate request held at 1	RTL	Retain; the counter it replaced was debug logic.

26	AHB default slave enabled; unmapped addresses return a bus error	RTL	Retain. Firmware that touched an unmapped address will now fault; confirm before sign-off.
27	Watchdog clock, enable and reset connected	RTL	Retain.
28	L2 data RAM model output-stage fix	FPGA model	FPGA-only model; no ASIC action. Confirm the foundry macro's read latency matches the CA53's expectation independently.
29	RT clock divide-by-610	FPGA-only	The ASIC has a real 32.768 kHz source; no action.
30	Flash I/O constraints, implementation-only, hold excluded on the data pins	XDC	Board-specific; no ASIC action. Revisit if the divider's enable structure changes.
31	Floorplan constraint (prepared)	FPGA-only	No ASIC action.
32	Boot ROM pin-mux ordering (MGSOC-49)	Firmware	Must be fixed before tape-out; hidden on FPGA.

13. Recommended next steps

Update, rev6: steps 1–3 were carried out in October (Sections 14–18); the floorplan (step 4) was not needed once build 42 met timing; the upstream fixes (step 5) were ported in the 9 October merge. Current recommendations are in Section 20.

1. Program the merged design and capture both UARTs (debug logger on PMOD3 IO[3], console on the UART0 pin) at 115200 8N1 across several auto-reset cycles. This is the first build with a trustworthy logger and the reclocked QSPI, and it answers the boot question directly.
2. If the console is silent, probe the SRAM bus rather than the peripheral window: the boot ROM's first write is to 0x0080_1ff0, followed by a 54-word burst to 0x80500c–0x8050dc. This separates "CPU never fetched" from "CPU ran but the peripheral path is dead".
3. Re-run the ARVALID reliability capture on the same build before treating it as the new baseline.
4. Enable the floorplan constraint (SLR0:SLR1, or a tighter region within them) as a single-variable experiment against the merged build, and check the clock-region count for the main clock drops.
5. Port the missing upstream fixes: the expansion-window counters and ID echo, and the two-stage interrupt synchroniser.
6. Fix the ROM/SRAM parity-lane widths, the one-hertz clock declaration, and the top-level reset structure; then make the testbench able to fail.
7. Decide which machine builds from here and keep one source tree.
8. Fold Section 12 into a tape-out sign-off process with a named owner. The register now has 32 items and no gate.

10.7 Found in October 2026 (builds 38–59)

#	Bug	Root cause	Fix
51	Boot restarted every ~300 µs forever; console never reached. The CPU fetched from ROM, ran enable_mmu, set up the stack and entered __rt_lib_init, whose first stack-based RET popped a zero link register.	FPGA SRAM wrapper connected the memory controller's active-low chip-enable and write-enable straight to the active-high block RAM, so every SRAM write was dropped and every read returned zero. The ROM wrapper inverts; the behavioural SRAM model is active-low, which is why simulation passed. The wrapper also aliased the RAM every 64 KB.	Inversions and address range corrected. Banner on the console on every boot since (10 October).
52	Boot stopped right after the banner; no bus traffic, no exception, no WFI. Present from the first banner (build 42) to build 53. Caches off, everything else worked.	Arm's TSMC example integration of the L2 tag RAMs (written for a 1 MB L2) feeds the RAMs the unmasked set index; this core has a 256 KB L2, and Arm's generic integration masks the index with l2_size. Tags were written and looked up at different RAM words; the first line fill through the L2 after enable_caches() corrupted execution. Only reached with a cache enabled. Found by bisection in RTL simulation of the FPGA design.	Arm's generic L2 tag integration with an FPGA variant of Arm's generic RAM model. Both caches verified on hardware, stock boot sequence end to end.
53	FPGA model of the L2 data RAM macro returned every read one cycle late.	The model had two register stages, from a misreading of the integration comment "two-cycle latency between the enable input and the data outputs"; Arm's generic model is a one-cycle registered read. Masked by row 52 and invisible with caches off.	One-cycle model; matches Arm's generic RAM over 4000 random cycles.
54	SPI_STARTUP has no branch for the on-board Winbond flash and the boot ROM jumps to the flash whether or not start-up succeeded.	Library supports Micron N25Q and SST26VF064B only; main() ignores the return value.	Firmware. A Winbond branch was written, validated step by step on hardware (QE via 50h/31h, 38h, C0h, EBh with mode A0; controller dummy count 6; a 10-clock all-ones transaction before FFh to leave continuous read) and supplied to SoC Labs.

55	Every 1-bit (SPI-mode) write from the QSPI controller reached the flash with the bits of each nibble reversed (0x02 arrived as 0x04).	The register block swaps the nibbles of each write byte for the 4-bit path; the 1-bit path then shifts each nibble out LSB first. In the upstream SoC Labs RTL.	Fixed in the project copy (MSB first within each nibble). Upstream fix pending.
56	Every 1-bit write from the QSPI controller carried one clock too many, so chip select rose off a byte boundary and the flash discarded all page programs and status-register writes. Erase (no data phase) worked, which made the chip look dead.	The data-out state exits on bit-counter value 7; reads use 6. In the upstream SoC Labs RTL. Never seen before because SPI_STARTUP writes only in 4-bit mode; a Winbond part needs a 1-bit status write before it can enter QPI.	Fixed in the project copy; unit simulation exact for 1-, 2- and 16-byte writes; QE set and page program verified on the chip. Upstream fix pending.
57	On-board flash declared locked or faulty: status register 1 read FF on every attempt.	The test command used opcode 09h, a typo for 05h; the line floated high. The chip was healthy throughout (rows 55–56 explain the failed writes).	Corrected; the chip programs and boots. Recorded as a misdiagnosis.
58	QSPI clock output has a zero-width glitch at the start of every transaction.	The reclocked controller muxes two flops that switch on the same clock edge. Introduced by the FPGA reclocking; the original clock-driven controller does not glitch.	Harmless on this board (the flash decodes correctly). FPGA-only; no ASIC action.
59	Any access to the flash window while XIP is inactive is a bus error, and the boot ROM reports every synchronous exception with the same message.	Interface design plus a one-line exception handler; the console could not tell a dead flash from invalid code.	Diagnostic ROMs print ESR/ELR/FAR. Firmware improvement recommended.
60	Monitor sessions were cut short: 12 of 17 blocks programmed, then reset.	The 10 s auto-reset in the FPGA top ran regardless of activity.	A falling edge on the console receive line suspends the auto-reset for about 215 s, renewed per byte. FPGA-only. Bench rule: after a crash, wait or press PB1.
61	Data abort on the first flash-window read after a control-register write.	Device-memory writes to the QSPI controller and the flash cache were not ordered against the following window load.	dsb sy after every control write in the boot software. Firmware.
62	Start-up code never runs scatter-loading: uninitialised globals are not zeroed and initialised data is never copied on silicon. Concrete case: the flash library's enter-QPI opcode, a global initialised to 0x38, lives in SRAM and holds whatever SRAM powers up with.	boot.s goes straight to __rt_lib_init and main; the RW/ZI region is a separate SRAM load region that only the simulation testbench preloads.	Firmware: call __main or copy/zero in boot.s; place RW in the flash load region. FPGA workaround: images zero-padded over ZI, RW preloaded over UART, the opcode written before every flash boot.

63	A system access to the CoreSight debug block (0x0130_0000) returns a bus error; dap_tests aborts.	Open; most likely the debug power domain is never powered up without a debugger.	Open, low priority. Simulation check of the debug power-up proposed.
64	SPI_WAIT_BUSY can hang after a short QSPI command.	It waits for BUSY to rise before waiting for it to fall; a 2-clock QPI command completes before the first status read.	Firmware: wait for not-busy only after a short delay (done in the diagnostic ROMs).
65	uart_tests cannot pass on hardware as written.	Receive buffer is a NULL pointer (handlers write into ROM space); the second half clears the console UART's transmit enable, muting every later printf; UART1 receive is enabled while the banner is still leaving UART0 (framing lost); the console is left at 2 Mbaud.	Test code. A corrected scratch copy passes both directions with interrupts and WFI on the loopback build.
66	The first UART-loaded program faulted at its second instruction; byte pairs 6–7 of each word and 32-bit stores at +4 landed one word up.	The new block-RAM slave advanced its write address on address accept even when no data beat was consumed; a first beat arriving after the address was stored at address+size.	Fixed; testbench extended with late-data cases. FPGA-only.
67	Two tests hung from flash while passing from RAM.	Their flash images were stale builds (July and January) with other UART clock assumptions; the packaging step reused any existing build.	Rebuilt; packaging now rebuilds unconditionally. Process.
68	First flash-window reads of one build did not set the FLASH_AHB access flag in the logger.	Open; minor.	Open.
69	A system access to the CoreSight debug block (0x0130_0000) aborts: dap_tests ESR_EL3 0x96000210, FAR 0x0130_0FD0. Closes row 63.	megasoc_cpu_ss.sv passes the full system address (paddr[30:2]) to the DAP-Lite system APB port. The DAP-Lite decodes its ROM table at debug-APB address 0 and forwards every other address to the Cortex-A53 debug APB, so the access reached the A53 at 0x30_0FD0 and got PSLVERR, reported as a synchronous external abort. Not a power problem: every debug power request was granted in simulation.	Offset within the 32 KB window on the port ({16'b0, paddr[14:2]}). RTL shared with the ASIC. Verified in VCS and on build 57 (dap_tests PASS). PR #5.
70	DBGROMADDR, the base the A53 reports in DBGDRAR/MDRAR for self-hosted debug, pointed at 0x1300_0000, unmapped.	28'h0013000 written for 0x0130_0000; the field counts 4 KB units.	28'h0001300. RTL shared with the ASIC. PR #5.

71	The DAP-Lite ROM table identifies nothing: designer and part zero, all 32 entries empty. dap_tests expected Arm part 0x4A1 and could never pass as integrated.	Arm's DapRomDefs.v is delivered unconfigured ('must be configured by the user'). It also shadows the A53's own ROM table at debug-APB address 0, and the 32 KB DEBUG window reaches only the ROM table (CPU0 debug sits at offset 0x10000).	Test identifies the block by the Component ID (0xB105100D, class-1 ROM table) and prints the PIDRs with a note (PR #5). Configuring the table and the window is an integration decision for SoC Labs; the Arm file is not in the public repository.
72	timer_tests reported 'INTSTATUS did not clear' from flash while passing from RAM (MBT-61).	test_timer_intstatus() leaves the timer running with RELOAD = 50 (2.5 µs) across INTCLEAR; fetched from the flash window with caches off, the INTSTATUS read lands after the timer has fired again. In cmsdk_apb_timer.v a set in the same cycle as a clear wins, correctly.	Test code: CTRL = 0 before the clear. Verified on builds 55 and 57. PR #7.
73	uart_tests hung at three different places on the board as its defects were removed one by one; six defects in all (MBT-59). Supersedes row 65.	(1) Loopback test commented out and not compiling: handler signatures, local char pointers shadowing the static buffers (row 65's 'NULL pointer'). (2) Results printed while UART1 RX, looped from UART0 TX, was still enabled at divider 10: an interrupt per console byte through an unbounded index, more than 600 interrupts, corruption. (3) WFI after each character: from flash the interrupt is taken before the WFI, which then sleeps for ever. (4) RX overrun test clears and re-checks the flag with two characters still in flight, then prints the error with UART0 TX disabled, where UartPutc() spins on TXBF for ever. (5) TX overrun test writes two bytes; a CMSDK UART moves the first into the shift register on the next PCLK, so two writes never overrun. (6) SWRESET sub-test spins for ever on the FPGA build (row 75).	Test code, all six: PR #6. Verified on build 57 from the physical flash (0 issues, TEST PASSED) and in VCS.
74	An 80-character console line loses its last character (MBT-62).	c_print_str() in system.c stops at 79 characters (i < 79); printf() takes that path for any format string without a % specifier. c_print() itself buffers 160.	Library code; one-line fix, not yet submitted.

75	system_sw_reset() never returns on the FPGA build; uart_tests' SWRESET sub-test spins.	The FPGA RTL snapshot has no megasoc_reset_ctrl (upstream logical/reset_controller); peripheral APB slot 11 is unconnected, so RESET_REQ is written into nothing.	Open. Port the upstream reset controller into the FPGA fork; meanwhile FPGA test builds skip the sub-test with -DSKIP_SWRESET_TEST.
76	Build 56's stock bootloader ROM printed at 625 kbaud; its boot was never observed.	The ROM was built in the upstream worktree, whose uart_stdout.c assumes 100 MHz and 3.125 Mbaud (divider 32); the board runs at 20 MHz.	Open. Corrected ROM built, not yet in a bitstream, so the stock boot path with the Winbond library code (PR #10) is still unproven. Process: build board ROMs from the board tree only.
77	megasoc_tech pins the AHB-QSPI submodule at a16f13e, the GitLab-era commit with the two 1-bit write bugs (rows 55, 56); the SoC Labs master (b6e9b76) has them fixed. The sdspi pointer 315fe083 does not exist on GitHub.	Submodule pointers not bumped when the controller was reworked upstream; an unpushed commit in sdio-controller.	Open. Bump both pointers before the ASIC netlist is taken; MBT-55/48 stay In Review until then.
78	The SoC Labs VCS flow (simulate/sim) aborts at the flash jump for every image, old or new.	The exception is taken by the ROM's handler, so the fetch at 0x400000 itself faults: a flash-model or ROM start-up mismatch in the simulation set-up, the symptom build 44 showed on the board.	Open. Flash-booted tests cannot be verified in that flow until it is fixed.
79	With the CPU caches on, a dirty cache line evicted from the L2 was written back to memory with the data of ANOTHER line of the same set; every block test larger than the L2 failed over the DDR4 (29 October: 4 MB block 28–68 % wrong, 64 KB eviction test 92 % wrong) although single transfers, cache-line fills, explicit clean+invalidate write-backs and narrow strobes were all correct. Every wrong word traced to the pattern of a line whose address differs only above bit 13.	A second instance of row 67 (N25). The Vivado fileset still contained Arm's TSMC 1 MB example integration of the L2 DATA RAMs (ca53_l2_datarams.v, RAM address A[13:0] unmasked) beside the generic one, and the tool compiled the example. The SCU addresses the data RAM with {set index, way}; for the 256 KB core the upper index bits are don't-care, so an unmasked RAM lets one line's data be written to one word and another line's read from it. Build 54 had replaced the TAG RAM integration only, and every RAM-mode run since build 47 had executed with the caches off (row 80), so nothing had exercised the data RAMs with evictions.	Build 59 (29 October): the generic, index-masked data and tag wrappers are the only definitions in the project (TSMC example and mem_integrations duplicates disabled; project file backed up). Verified 30 October: ddr4_fpga_tests v8 0 errors from DDR4 and from flash with the caches on; 310 block RAM primitives instead of 522 (the constant mask lets Vivado trim the RAMs to 256 KB).

80	Every RAM-mode test run of builds 47–58 executed with the CPU caches off (SCTLR_EL3 0x40c52839: M=1, C=0, I=0 at test entry in both boot modes); the DRAM port had only ever seen single 64-bit transfers, and row 79 stayed hidden for two weeks.	The monitor ROM leaves the caches off, and the SoC Labs library's boot.s calls enable_mmu() only; enable_caches() is called by the stock bootloader, not by the applications. The flash-booted tests of builds 51–57 ran the same way after the monitor's jump.	ddr4_fpga_tests enables the caches itself (after setting SMPEN, row 81) and prints the state it found. Proposal for SoC Labs: enable the caches in the library start-up, or in the monitor before the jump, so every test runs as the silicon will.
81	The library's enable_caches() never sets CPUECTLR.SMPEN, which the Cortex-A53 requires before the data cache is enabled; coherency between L1 and L2 is undefined without it.	cpu_asm_codes.c: SCTLR |= 0x1004 only. CPUECTLR read 0 before the test set bit 6. (The same-set corruption of row 79 was briefly attributed to this; the cache-resident test T5 cleared it.)	The test writes SMPEN (msr s3_1_c15_c2_1) before enabling the caches. Arm-licensed file: proposal to SoC Labs, not published.
82	No stock test can access DRAM above 4 GB although the library's page table describes 512 GB; the flash-mode DDR4 test took a level-0 translation fault at 0x1_0000_0000 (ESR_EL3 0x96000044).	enable_mmu() programs TCR_EL3 = 0x00102F20: T0SZ 32 (4 GB of VA) and PS 0 (32-bit PA). The DRAM window is 8 GB (0x8000_0000–0x2_7FFF_FFFF).	The test widens TCR_EL3 to T0SZ 25 / PS 0b010 (39-bit VA, 40-bit PA = the A53's PARange) with ISB, TLBI ALLE3 and DSB before its first access. Arm-licensed file: proposal only.
83	Code booted through the monitor's Winbond 'n' path ran about 30 times slower than the same code loaded with 'g' into DDR4 (T1 330.8 M against 0.77 M counter ticks); the suite reported HANG during the 4 MB block test.	winbond_startup()'s jump omits the CG092 flash-cache enable (cache_on()) that the 'g' path performs, so every instruction fetch was an uncached QSPI read. The generic counter runs at 20 MHz with CNTFRQ_EL0 unset.	Not changed: suite --timeout added, per-test CNTPCT times in the test. Proposal: call cache_on() before the Winbond jump in the monitor.
84	SoC Labs' own FPGA L2 data RAM wrapper (mem_integrations/FPGA/ca53_l2_datarams.v) provides 64 KB of storage for the 256 KB L2 and ignores the way bits.	ADDR(l2_dataram_addr_i[11:2]) into cmsdk_fpga_sram with AW = 12: 1024 words of 64 bits per instance, 16 instances; address bits [1:0] (the way) are not connected.	Not used by this build (disabled with the other duplicates in build 59). Reported to SoC Labs for the record.

85	Remapping the on-chip SRAM Normal cacheable raised an SError (ESR_EL3 0xBF000000, EC 0x2F, IDS) at the first cached store to 0x81_0000 (ddr4_fpga_tests T7, version 7).	A test address fault with a design finding behind it. The NIC-400 decodes only 64 KB of RAM at 0x80_0000 (asib_A53 decode_int[2]: 0x800000–0x80FFFF; Socrates memory_map.xml range 0x10000) while the FPGA SRAM wrapper holds 128 KB in two 64 KB banks. The write-allocate line fill to the unmapped address returned DECERR from the default slave, reported asynchronously because the store had already retired; as Device memory the same access is a synchronous abort. The SIE-300 controller cannot have produced the error: its gate-response input is tied to stall.	Test v8 uses 0x808000–0x80FFFF and passes (32 KB of dirty SRAM lines evicted through the L2 and verified). The unreachable upper 64 KB and the absence of a RAM size symbol in sys_memory_map.h are reported to SoC Labs (ticket proposed, Low).

14. From silence to the banner — the boot loop (1–10 October, builds 38–42)

Rev5 closed with a trustworthy debug logger, a reclocked QSPI controller and a boot console that had never been observed. The first October builds put the merged design on the board. Build 38 ran the whole-bus logger on the merged tree; builds 39–41 added sticky milestone flags (first fetch, first ROM data, first SRAM write, the boot software's first stack push at 0x801BF0, first flash access, first UART register write, first UART byte) and a field that samples the last accepted CPU address every 50 ms.

14.1 What the logger showed

Every boot was identical: the CPU fetched from ROM, wrote to SRAM at the expected address, never fetched the exception vectors, never reached the UART, and the sampled fetch addresses kept returning to the reset path. The boot restarted about every 300 µs with no warm reset and no exception. That pattern — a successful stack push followed by a return to the reset vector — points at a stack read returning zero: the first stack-based RET in __rt_lib_init popped a zero link register and jumped to address 0.

14.2 Root cause: the FPGA SRAM wrapper's polarity

The memory controller drives its chip-enable and write-enable active low. The FPGA block RAM behind it has active-high CS and WREN. The wrapper written in July connected them without inversion, so during every access CS was 0: writes were dropped and the read data was forced to zero. The ROM wrapper inverts the same signals, which is why ROM reads had always worked, and the behavioural SRAM model used in simulation is active low, which is why every simulation passed. The wrapper also indexed the RAM by the wrong address bits, aliasing every 64 KB. Both were corrected and checked in a unit simulation with the controller's control polarity (bug 51).

14.3 Build 42: the first firmware output on this board

Build 42 (9 October, also carrying the Windows Build G merge that closed timing: WNS +21.5 ns, WHS +0.01 ns, one clock domain) printed ***SoCLabs MegaSoC*** on the console at every 10 s auto-reset. It then stopped: no bus traffic for the rest of the window, no exception, no WFI. The next statement in bootloader.c after the banner is enable_caches(). Build 43 (12 October) used a scratch boot ROM with the two cache-enable calls removed and ran straight through to the flash start-up, proving that cache enable was the stop (bug 52, resolved in Section 18) and separating it from everything that followed.

Rule adopted here and kept for the rest of the period: the FPGA boot ROM is generated from a scratch copy of the boot software for each diagnostic build and installed over bootrom.v; the original is kept beside it (bootrom.v.orig_20260812fw) and must be restored for any non-diagnostic build. Every hardware observation records which ROM was in the bitstream.

15. The jump to flash, and a monitor in the boot ROM (12–14 October, builds 43–46)

15.1 What the jump hit

With caches off, build 43 reached ***Flash Enabled…Booting*** and took a synchronous exception at the jump to 0x400000. The flash window returns a bus error for any access while execute-in-place is inactive, and the boot software ignores SPI_STARTUP's return value and jumps regardless (bug 54), while its exception handler prints one line for every cause (bug 59). Build 44's diagnostic ROM printed the controller registers, the flash's JEDEC identity and the exception syndrome: the chip answered EF 60 16 — a Winbond W25Q32FW, a part SPI_STARTUP does not support — the start-up took the Micron path and left the controller in 4-bit mode against a chip still in 1-bit mode, the window read 0xCCCCCCCC, and the fetch faulted as an undefined instruction. That proved the whole path CPU → interconnect → flash cache → QSPI controller → pads → flash die on the first attempt.

15.2 The monitor ROM and the host tool

Builds 45 and 46 replaced the boot ROM with an interactive monitor (UART command set: memory read/write/dump, raw QSPI commands, JEDEC, erase, program with checksum, memory load, checksum verify, jump, and later production-boot variants) and a Python host tool, haps_qspi.py, that drives it over the console port with termios only. Build 46 added a hold-off of the 10 s auto-reset on console activity (bug 60) and a boot software ROM with barriers after every control write (bug 61). UART0 receive was proven here for the first time.

15.3 The controller under unit simulation

To separate controller from chip, the QSPI register block, controller, clock divider and (later) the AHB interface were simulated with xsim against a scripted flash. This found the nibble bit-order defect in 1-bit writes (bug 55) and the SCLK start glitch (bug 58), and confirmed reads exact. Because the clock count was measured in rising edges rather than real pulses, the extra-clock defect (bug 56) was missed until 19 October.

15.4 The flash declared dead — and why that was wrong

On 14 October the chip answered every identification read but no array read, and ignored every write, erase result notwithstanding; status register 1 read FF constantly, unchanged across board power cycles. It was declared locked or faulty and the work moved to plan B (Section 16). On 19 October the status read was found to use opcode 09h, a typo for 05h (bug 57), and the write failures were traced to the controller (bugs 55–56). The chip was healthy throughout. The misdiagnosis cost no board time — plan B was needed anyway for RAM-resident tests — but it is recorded in full.

16. Running the SoC Labs tests from RAM over UART (14–15 October, builds 47–48)

Update, Revision 8: every RAM-mode run described here and in later sections, up to build 58, executed with the CPU caches off — the monitor ROM leaves them off and the library's boot.s enables

the MMU only (row 80, Section 29.2). The results stand; what they exercised was single 64-bit transfers on the DRAM port.

16.1 A block-RAM slave on the DRAM port

The interconnect's DRAM target (0x8000_0000, mapped Normal, cacheable, executable by the boot page tables) had an error slave behind it. Build 47 replaced it with a 512 KB AXI4 block-RAM slave written for the purpose (one array per byte lane, because Vivado caps a single inferred array at 1 Mbit; INCR, WRAP and FIXED bursts; one outstanding read and write; unit testbench). Build 48 fixed a first-beat address bug found on the board (bug 66), the testbench now covering data beats arriving after the address.

16.2 Images and loading

A new scatter file places a test's code and data in one region at 0x8000_0000 with stack and heap in SRAM; fromelf produces a flat image, zero-padded over the uninitialised-data region because the start-up code never zeroes it (bug 62). The host tool loads the image through the monitor in 4 KB blocks with checksums and jumps to it. The runner mode (suite) does this for a list of images unattended, waits silently through the board's hold-off between tests, follows a console speed change if a test makes one, and writes a summary and per-test logs.

16.3 Results

15 October, build 48: hello_world ran to ** TEST PASSED ** — the first SoC Labs test completed on this board. The full set the same day:

Test	Result from RAM	Note
hello_world, mem_tests, timer_tests, neon_tests, peripheral_test	PASS	
gic_tests	PASS	Timer interrupts through the GIC-400 to the core and its handler: first hardware evidence for the interrupt path and the two-stage synchroniser.
low_power_tests	PASS	Core retention with wake by timer interrupt; ROM power domain switched off and on through its PPU: validates the Q-channel and PPU fixes on the board.
dap_tests	FAIL (abort)	Debug block read returns a bus error (bug 63).
sdio_tests	FAIL (abort)	No card fitted.
pl011_uart_tests	FAIL	Needs a transmit-to-receive loopback that only the testbench provides (Section 17.5).
uart_tests	HANG	Needs UART0/UART1 cross-wiring (Section 17.5) and has four bugs of its own (bug 65).

16.4 A silicon-relevant finding: scatter-loading never runs

Reading the disassembly to understand why consecutive tests could inherit each other's RAM contents showed that the SoC Labs start-up code never calls the scatter-loader: uninitialised data is never zeroed and initialised data in SRAM is never set on silicon. The simulation testbench hides

both. The concrete case in the boot path is the flash library's enter-QPI opcode, a global initialised to 0x38 but living in uninitialised SRAM, so a flash boot with an SST part would send a random command (bug 62, N35).

17. The flash path: emulator, controller bugs, physical flash (15–20 October, builds 49–52)

17.1 An in-FPGA flash emulator (build 49)

With the chip believed dead, build 49 placed an SST26VF064B-compatible flash model on the controller's pad-side signals inside the technology wrapper, backed by a second 512 KB window of the RAM slave reachable over UART. Everything on the SoC side of the pins — controller, flash cache in bypass, XIP window, the unmodified SPI_STARTUP taking its tested SST branch — was then exercised without a working chip. The emulator was first simulated against the real controller stack (zero errors), then a RAM-resident probe on the board read the whole image back through the window with 8-, 32-, 64- and 128-bit loads, including line-crossing ones, with zero errors. All seven passing tests passed from the emulated flash. Switchable with one parameter (USE_FLASH_EMU, now 0).

17.2 The controller's 1-bit write path (build 50)

A review of the flash evidence on 19 October found the status-read typo (bug 57) and prompted a count of real clock pulses per transaction in the controller simulation: 17 instead of 16 for the QE status write, 161 instead of 160 for a 16-byte page program (bug 56), with the data decoded as 0x04 for 0x02 (bug 55). Both defects were reproduced on SoC Labs' original, clock-driven controller RTL, so they are upstream and shared with the ASIC; they never showed because SPI_STARTUP writes only in 4-bit mode. Build 50 carried the fixes and switched back to the real pins. On the board the QE write succeeded at the first attempt and a 16-byte page program read back exactly: the first data ever programmed into the chip from the SoC.

17.3 The Winbond start-up, validated step by step

Register-level sessions through the monitor then validated the start-up sequence drafted for SoC Labs: 38h accepted (a 1-bit JEDEC read afterwards returns FF, as a chip in QPI mode should), JEDEC in 4-bit mode returns EF 60 16 (all four data lines work), C0h, 0Bh and EBh reads exact, the XIP window in continuous-read mode returns the data. Two corrections to the draft came out of it: the controller's dummy count for EBh is 6, not 8, because the chip counts the two mode-byte clocks within its eight; and leaving continuous-read mode needs a 10-clock all-ones transaction before FFh, since 2-clock commands are taken as address nibbles. A third gotcha: the software reset in the exit sequence clears the volatile QE bit, so QE must be set again before every 38h. hello_world booted from the physical flash by hand on 19 October, with the flash cache enabled by the jump.

17.4 Builds 51 and 52: automation and loopbacks

Build 51's monitor ROM runs the corrected Winbond start-up at every reset and boots whatever image is in the flash; the runner's --winbond mode programs each test into the chip over UART

(erase 1.3 s, program 7 s for 68 KB, verified through the 1-bit window), preloads the SRAM data and boots. 20 October: 7 of 7 from the physical flash, dummy count 6 on the first attempt every time.

17.5 UART loopbacks and the uart_tests bugs (build 52)

The two UART tests rely on wiring the simulation testbench provides: UART0 transmit to UART1 receive and back, and PL011 transmit to receive. Build 52 adds the same under the FPGA-only define; UART0's receive is the AND of the host line and UART1's transmit, both idle high and never active together, so the console and the loader keep working. pl011_uart_tests then passed from the real flash. uart_tests still failed, and probes (polled transfers both ways, and the same string sent five ways) showed the cross-wiring and both receive interrupts working while the test itself has four defects (bug 65): a NULL receive buffer, the console transmitter disabled in its second half, the second receiver enabled while the banner is still shifting out, and a 2 Mbaud console. A scratch copy with those corrected passes both directions with interrupts and WFI.

18. The cache-enable stop (20–21 October, builds 53–54)

Update, Revision 8: build 54 corrected the integration of the L2 TAG RAMs only. The L2 DATA RAMs kept Arm's TSMC 1 MB example wrapper (the same unmasked set index) until build 59; the stock boot and hello_world ran correctly on builds 54–57 because their working set never evicted dirty lines of two same-set addresses. The defect surfaced on 29 October with a 4 MB block test on the DDR4 (row 79, Section 29).

18.1 First attempt: the L2 data RAM model

The FPGA models of the Cortex-A53's TSMC cache RAM macros are the only part of the core that is not Arm's RTL, so they were the first suspect. The L2 data RAM model had a two-register read pipeline where Arm's generic model has one (bug 53); a unit comparison over 4000 random cycles showed 1370 mismatches against zero for the corrected model. Build 53 carried the fix. On the board the stop was unchanged.

18.2 Simulating the FPGA design itself

Rather than guess further, the exact Vivado file list (769 files, exported from the project) was compiled in VCS with a 20 MHz differential clock, the reset button, pulled-up flash pins and a testbench that prints every byte the software writes to UART0's data register. A boot ROM built from a stepwise probe — instruction cache on, off, data cache on with memory tests, both on — ran in 45 s of CPU time per 400 ms of SoC time and reproduced the board exactly: the console stopped the moment the instruction cache was enabled. With the simulated RAMs preset to zero, as FPGA block RAM powers up, the match with hardware was complete.

18.3 Bisection and root cause

Swapping the cache RAM models for Arm's generic ones made the simulation pass. All eleven FPGA macro models matched Arm's generic RAM cycle for cycle in isolation, and presetting the tag RAMs to all ones (Arm's generic model does so; an all-zero tag reads as valid) changed nothing, so the difference was in the integration files, not the models. Swapping them one at a time showed that only the L2 tag RAM integration decides pass or hang. Arm's TSMC example for this macro library is

written for a 1 MB L2 and feeds the tag RAMs the unmasked set index; this core is configured with a 256 KB L2, and Arm's generic integration masks the index with l2_size. Tags were written and looked up at different RAM words, so the first instruction-cache line fill through the L2 after enable_caches() corrupted execution (bug 52). The same file had been in every build since 42. It is FPGA-only: silicon uses SoC Labs' own 256 KB integration.

Build 54 uses Arm's generic L2 tag integration with an FPGA variant of Arm's generic RAM model (defined power-up contents, output held through non-read cycles, synthesisable). The full stepwise program passed in simulation first — instruction cache, data cache, both, 64 KB memory tests at zero errors — then the build was launched. One false alarm along the way is recorded: a data-cache test appeared to freeze in simulation until the simulated RAM slave was zero-initialised; never-written words had read back as undefined values.

18.4 Confirmation on hardware, 21 October

Program (RAM-resident, build 54)	Result
cache_steps: IC IALLU; SCTLR.I on + code loop; I off; C on + 1 KB SRAM + 64 KB RAM; C off; C and I on + 64 KB RAM	PASS, 0 errors in every memory test
cache_probe: the stock enable_caches() of bootloader.c, 8 KB SRAM, two 64 KB RAM tests, code loop, disable, retest	PASS
cache_flashboot: enable_caches at EL3 and EL1 as bootloader.c, Winbond start-up, jump to the image in the physical flash	PASS — Hello SoCLabs MegaSoC, TEST PASSED with both caches on
Regression: seven tests from the physical flash (suite --winbond)	7 of 7 PASS

The stock boot sequence — reset, boot ROM, flash start-up, cache enable, jump into the XIP window — now runs end to end on the board.

19. Current status (21 October 2026) — supersedes Section 11

Update, rev7: this section is the position on 21 October and is superseded by Section 25.

Area	Status	Evidence
CPU first fetch, ROM, SRAM, boot console	DONE, on hardware	Banner every boot since build 42 (Section 14).
UART0 both directions; UART1 and PL011 loopbacks	DONE, on hardware	Monitor and loader; pl011_uart_tests PASS; corrected uart_tests PASS (Section 17.5).
Interrupts through the GIC-400; PPU power sequencing	DONE, on hardware	gic_tests and low_power_tests PASS from RAM and flash (Section 16.3).
QSPI controller, flash cache, XIP window	DONE, on hardware	Emulated and physical flash; probe at all access widths (Section 17).
Physical on-board flash: program, boot, execute	DONE, on hardware	7 of 7 tests from the chip (builds 51, 54).
CPU caches; stock boot sequence	DONE, on hardware	Build 54, Section 18.4.
Timing	DONE	All constraints met since build 42; build 54 WNS +14.07 ns, WHS +0.01 ns, 0 critical warnings, 0 latches, one clock domain.
dap_tests (debug block read)	OPEN, low	Bug 63 / N36.
sdio_tests	N/A	No card fitted.
Upstream items for SoC Labs	OPEN	Bugs 54, 55, 56, 62, 64, 65 and the TSMC L2 example integration note (Section 20).
Jira	OPEN	Atlassian unreachable from the server since 14 October; all updates paste-ready in the Desktop log.

19.1 Test matrix, build 54

Test	From RAM (UART)	From emulated flash	From physical flash	Note
hello_world	PASS	PASS	PASS (also with both caches on)	
mem_tests	PASS	PASS	PASS	
timer_tests	PASS	PASS	PASS	
neon_tests	PASS	PASS	PASS	
peripheral_test	PASS	PASS	PASS	
gic_tests	PASS	PASS	PASS	
low_power_tests	PASS	PASS	PASS	
pl011_uart_tests	—	—	PASS	Build 52 loopback.
uart_tests	PASS (corrected copy)	—	HANG (as written)	Test bugs, bug 65.
dap_tests	FAIL	—	—	Bug 63.
sdio_tests	FAIL	—	—	No card.
coremark, dma350_tests	—	—	—	Do not build upstream.

19.2 Timing progression, October builds

Build	WNS (ns)	WHS (ns)	Note
42	+21.5	+0.01	Build G merge; first banner
48	+19.447	+0.009	512 KB RAM slave
49	+29.32	+0.01	Flash emulator
50 / 51	+15.067	+0.010	Controller write fixes; ROM v5.2
52	+20.349	+0.010	UART loopbacks
53	+16.332	+0.010	L2 data RAM model
54	+14.072	+0.010	L2 tag integration; 743 → 703 block RAM tiles

20. Tape-out revert register additions, and recommended next steps

Items 1–32 (Sections 12 and the earlier revisions) still apply with the changes noted. New items from this period:

#	Item	Type	Action before tape-out
33	SRAM wrapper polarity and address range (SRAM/fpga/SRAM_wrapper.sv)	FPGA-only	No ASIC action; the behavioural and ASIC SRAM paths are correct.
34	bootrom.v holds a monitor/diagnostic ROM (v5.2)	FPGA image	Restore bootrom.v.orig_20260812fw, regenerated from the current bootloader, for any deliverable build.
35	512 KB block-RAM slave on the DRAM port; flash emulator (USE_FLASH_EMU = 0) and its backdoor	FPGA-only RTL	No ASIC action; remove or keep behind the define.
36	QSPI controller 1-bit write fixes (data phase ends on the byte boundary; MSB first within nibbles)	RTL, shared with the ASIC	Must go upstream and into the ASIC netlist; otherwise no 1-bit status write or page program works from the SoC.
37	UART0/UART1 and PL011 loopbacks under FPGA_BUILD	FPGA-only	No ASIC action.
38	Cache RAM FPGA models (defined power-up, hold through writes, 1-cycle L2 data); Arm generic L2 tag integration with ca53_generic_ram_fpga.v	FPGA-only	No ASIC action; the ASIC uses SoC Labs' mem_integrations. Note for any other FPGA port of this core: Arm's TSMC example L2 tag integration is for a 1 MB L2 only.
39	Console-activity hold-off of the auto-reset; whole-bus debug logger taps	FPGA-only	No ASIC action.
40	Winbond branch for SPI_STARTUP (dummy count 6; 10-clock exit); return value checked before the jump	Firmware	Adopt upstream if a Winbond part is a target; at minimum check the return value.
41	Scatter-loading never runs; SPI_WAIT_BUSY race; uart_tests defects	Firmware / test code	Fix upstream before silicon bring-up (bugs 62, 64, 65).

20.1 Recommended next steps

Update, rev7: items 1, 2 and 4 are done or overtaken (Sections 21, 22 and 24); the list is superseded by Section 26.3.

1. Send SoC Labs the two controller fixes with the simulation evidence (pulse counts and decoded bytes on the original RTL), the Winbond branch with its two corrections, and the firmware and test-code items; the package on the Desktop (for_daniel_flash) holds the first two groups.
2. Paste the October Jira updates (MBT-23…36 status changes; N24–N40) when Atlassian is reachable; close MBT-26 with the real root cause (L2 tag integration, not the data RAM model).
3. Decide the deliverable ROM: restore the original boot ROM, or adopt the Winbond-capable start-up, and rebuild once for the record.
4. Low priority: simulate the debug power-up to settle dap_tests (bug 63); try a card for sdio_tests.
5. Keep the RTL simulation of the FPGA design (scratchpad n25/sim: files.f from the Vivado list, tb_haps.sv) as the first tool for any future FPGA anomaly; it reproduces the board at 45 s per 400 ms and found the cache stop in one night after a week of hardware-only probing.

21. Upstream contribution and the move to GitHub (21–23 October, build 55)

21.1 What went upstream

On 21 October the bring-up was packaged for SoC Labs and pushed to the public GitHub organisation as two pull requests, both reviewed and merged by Daniel Newbrook on 22 October. MegaSoC-Project PR #7 (86 files) adds fpga/targets/haps_sx/bringup/: the FPGA top and its constraints, the whole-bus debug UART logger, the 512 KB block-RAM slave on the DRAM port, the flash emulator, the clock-gate bypass, the behavioural models of the TSMC cache RAM macros, the build chain with its constraint guards, the VCS testbench of the FPGA top, the host tools (monitor driver, capture, logger decoder), the monitor ROM and test images, diffs of every SoC Labs RTL change against the GitLab base this work started from, and the documentation (Revision 6 of this record, the bug log with Jira cross-references, the per-build notes). No Arm-licensed source is included; the README lists what to take from the Arm IP environment and the local edits to make. The same PR teaches flows/makefile.software to accept a single-file hex image, the form fromelf produces once RW/ZI live inside the ROM load region. MegaSoC-Tech PR #4 carries the two firmware fixes of Section 16.4 and bug 62: scatter-loading now runs at start-up (MBT-56 — RW/ZI are execution regions of the ROM load region, and the master core enters the C library through __main) and the weak abort handler prints ESR_EL3, ELR_EL3 and FAR_EL3 (MBT-50).

The two QSPI controller write bugs (rows 55 and 56; MBT-55 and MBT-48) needed no change from us after all. An end-to-end simulation of the current AHB-QSPI master (b6e9b76, 146 commits after our GitLab base) — register block, mux, controller, divider and AHB interface — gives the QE status write as 16 clocks carrying 0x02, the two-byte status write as 24 clocks carrying 00 02 and a 16-byte page program as 160 clocks with the data exact. The controller was reworked there: write data loaded once, a 5-bit byte compare, the SPI-mode data transform moved into the register block. Our fork was simply stale. What remains is that megasoc_tech still points at the old commit (row 77); the fixes reach the ASIC netlist only when that pointer is bumped.

One finding from the same day concerns the SoC Labs simulation flow rather than the design: a control run of hello_world in simulate/sim aborted at the flash jump exactly as the July mem_tests log had, old image or new. The exception is taken by the ROM's own handler, so the fetch at 0x400000 itself faults — a flash-model or ROM start-up mismatch in the simulation set-up, the same symptom build 44 showed on the board. Until it is fixed, flash-booted tests cannot be verified in that flow (row 78).

21.2 The work folder on GitHub

On 22 October, once the pull requests had merged, the local tree stopped being a GitLab clone. A full backup was taken first (~/backup_megasoc_project_20261021: a 20 GB archive plus git metadata and patches); megasoc_project main was then set to the GitHub main (53a561d, PR #7 merged) and every submodule to its public counterpart. The FPGA work area FPGA_Tasos/ is ignored by git and stays on the server only, because it holds Arm IP. megasoc_tech carries one local branch, local/haps-fpga, with the board-only edits that must never go upstream: the console divider for a 20 MHz clock and 115200 baud, and the scratch test directories. Syncing exposed the two submodule pointer

problems of row 77: the AHB-QSPI pointer still at the buggy GitLab-era commit, and an sdspi pointer naming a commit that was never pushed to GitHub.

21.3 Build 55 and the first regression from the GitHub tree (23 October)

Build 55 rebuilt build 54's sources unchanged from the moved folder, to prove the FPGA flow independent of the git move: WNS +15.130 ns, WHS +0.009 ns, no critical warnings. The firmware was then rebuilt from the updated tree as single-load-region images, which boot from the flash with no SRAM preload at all — the first time the whole regression ran that way, so MBT-56 is confirmed for every test and not only for hello_world. Seven tests and pl011_uart_tests passed from the physical flash. timer_tests, which upstream had extended with a new INTSTATUS sub-test since our fork, failed with 'INTSTATUS did not clear' (Section 21.4). dap_tests aborted as before, but the merged handler now gave the syndrome in full — EC 0x25, ISS 0x210, FAR 0x0130_0FD0 — the detail that pointed Section 22 at the address rather than at power. uart_tests hung as in Revision 6.

21.4 The timer test race (MBT-61)

The new sub-test leaves the timer running with RELOAD = 50, 2.5 µs at 20 MHz, writes INTCLEAR and reads INTSTATUS expecting it clear. From the flash window with the caches off a 16-byte line takes about 20 µs to fetch, so the read lands after the timer has fired again; in cmsdk_apb_timer.v a set in the same cycle as a clear wins and the timer keeps reloading, which is correct behaviour. The same binary passes from RAM, and passes in simulation by margin. Stopping the timer before the clear makes the check independent of fetch speed; the fixed image passed from the flash on builds 55 and 57 (PR #7, Jira MBT-61, raised 28 October).

22. The debug block: root cause of dap_tests (24–27 October, builds 56–57)

22.1 Reading the syndrome

dap_tests had failed on every build since the tests first ran from RAM (row 63), and Revision 6 recorded the likeliest cause as the debug power domain never being powered without a debugger attached. The merged abort handler changed the question. A synchronous external abort on a data access (EC 0x25, ISS 0x210) at FAR 0x0130_0FD0 is DAP_DBG_BASE plus 0xFD0 — the first ROM-table identification register the test reads. An external abort at a precise address is a bus answer, not a power state.

22.2 Simulation of the FPGA design, second use

The VCS build of the exact Vivado file list from Section 18.2 was reused with a boot ROM made from dap_tests and monitors on the DAP-Lite's ports and on the Cortex-A53 debug APB. It reproduced the board's syndrome exactly and showed the mechanism in one run. The NIC-400's DEBUG target is the 32 KB window 0x0130_0000–0x0130_7FFF (decode entry 6 of nic400_asib_A53_decode_megasoc_main.v), and the interconnect presents the full system address on its APB port. megasoc_cpu_ss.sv wires that address straight to the DAP-Lite's system APB address input as paddr[30:2]. The DAP-Lite decodes its ROM table at debug-APB address 0x0000_0xxx (ROMADDR in DAPLiteDefs.v) and forwards every other address to the A53's 20-bit debug APB, so

0x0130_0FD0 arrived at the core as 0x30_0FD0, an address with nothing behind it; the core returned PSLVERR, and the bridge reports PSLVERR as an external abort. The debug power inputs (DEVICEEN, nCDBGPWRDN, nCSOCPWRDN, CDBGPWRUPACK) were asserted throughout; power was never the problem.

One wrong turn is recorded. The first fix presented paddr[21:2], and the simulation changed nothing — because bits 21 and 20 of the base address 0x0130_0000 are set, so the forwarded address was still outside the ROM table. The port needs the offset within the window, paddr[14:2] zero-extended; with it all twelve ROM-table reads complete with PSLVERR low.

22.3 What the ROM table contains, and what the test should check

The test then failed differently: it compared the peripheral identification registers against Arm's values for a configured DAP (part 0x4A1), and DapRomDefs.v as delivered — an Arm file marked 'must be configured by the user' — has designer and part number zero (PIDR 00 00 08 00 / 00 00 00 00) and all 32 entries empty. The Component ID registers are fixed by the architecture, 0xB105100D for a class-1 CoreSight ROM table, so the test now identifies the block by those and prints the PIDRs with a note when they are unconfigured. A third defect was found alongside: DBGROMADDR, which feeds the A53's DBGDRAR and MDRAR registers for self-hosted debug, was 28'h0013000 (0x1300_0000, unmapped) in place of 28'h0001300 (0x0130_0000). Three items are for SoC Labs' integration rather than for this branch: DapRomDefs.v should be configured (designer Arm, a part number, entries for the A53 debug components); the A53's own ROM table at debug-APB address 0 is shadowed by the DAP-Lite's; and the 32 KB DEBUG window reaches only the ROM table, while CPU0's debug registers sit at offset 0x10000. The Arm files live in the private IP library, not on GitHub, and were not edited.

22.4 Builds 56 and 57, and the confirmation

Build 56 carried the two cpu_ss changes with the stock SoC Labs bootloader ROM in place of the monitor, so that the stock boot path with the new Winbond start-up (the library code of PR #10) could be proven at the same time; a two-bitstream procedure programmed the flash through build 55's monitor and booted with build 56. Its console produced nothing readable. The ROM had been built in the upstream worktree, whose uart_stdout.c assumes a 100 MHz clock and 3.125 Mbaud — at 20 MHz that divider gives a 625 kbaud console (row 76). A corrected ROM was built but not put into a bitstream; the stock boot path with that code is therefore still unobserved and is flagged as such in PR #10. Build 57 (26 October; WNS +14.240 ns, WHS +0.009 ns) is the same RTL with the monitor ROM v5.2, so the usual flow works. On 27 October dap_tests from the physical flash printed CIDR = 0xb105100d (class 1), the expected note about the unconfigured designer and part fields, and TEST PASSED; hello_world and then the seven-test regression passed on the same build with the fixed timer_tests image. MBT-58 was closed on hardware 23 days after the abort was first seen.

23. uart_tests: six defects, and a passing test (26–27 October, build 57)

23.1 Where Revision 6 left it

Row 65 recorded four defects from probes on build 52, and a scratch copy that passed the two loopback directions with interrupts and WFI. Running the real test from the flash — loopback test re-enabled, the rest unmodified — showed that the picture was larger. Three of the six defects are timing races that neither a simulation from ROM nor a program loaded into RAM shows; they appear with code fetched from the flash window, which is also how the test will run on silicon.

23.2 Image by image

The first image hung after printing 'UART0 TX --> UART1 RX'. UART1's receiver and its interrupt had stayed enabled while the results were printed: with UART0 transmit looped into UART1 receive, every console byte, sent at 115200 and received at divider 10, raised a receive interrupt, and the handler stored each one through an unbounded index — more than 600 interrupts, memory corruption, hang. The fix stops the receiver and its GIC interrupt before printing, bounds the handlers to the 64-byte buffers and restores the console divider before the results.

The second image hung in the other direction. Each character was followed by a wait-for-interrupt instruction. With code fetched from flash the call to the wait routine is itself a line fill, long enough for the receive interrupt to arrive before the instruction is reached; a WFI entered after its interrupt has already been taken sleeps for ever. The first direction had won the same race by chance, and the simulation, running from ROM, never saw it. The fix is a bounded poll on the index the handler increments; reception stays interrupt-driven and a timeout reports an error instead of hanging.

The third image passed both loopbacks and hung in the RX overrun test. That test sends 32 bytes from UART1 into UART0 without reading them, then checks, clears and re-checks the overrun flag. It waits only for UART1's holding register, so two characters are still in flight when it clears; a late one re-sets the flag, the error path is taken, and the message is printed with UART0's transmitter disabled — where the console putc routine spins on TXBF for ever. The simulation won that race from ROM and reached the next test, where it found the fifth defect: the TX overrun test writes two bytes and expects an overrun, but a CMSDK UART moves the first byte from the holding register into the shift register on the very next clock when the transmitter is idle, so the second write always finds the register free. A third write while the second byte waits is what overruns. The sixth defect is environmental: the SWRESET sub-test writes RESET_REQ and waits for the reset, and the FPGA RTL snapshot has no megasoc_reset_ctrl (row 75), so it waits for ever; FPGA test builds skip it with a define and it stays enabled upstream.

Two smaller things came out alongside. A plain 80-character line lost its newline on the board and in simulation because c_print_str() stops at 79 characters (row 74, MBT-62). And the 'divider 10 = 2 Mbaud' of Revision 6 is wrong: the CMSDK UART's divider has a minimum of 16, and with 10 the baud counter reloads every clock, so both UARTs run at PCLK/16 = 1.25 Mbaud in those tests. The console adapter follows neither rate, and the runner option written for the 2 Mbaud assumption (--baud-after) must not be used with the corrected test, which prints everything at the console rate.

23.3 The UART testbench

The dap testbench was extended for this work. It prints every byte written to UART0's data register, so the output is readable whatever the divider; it logs UART0 CTRL and BAUDDIV writes, UART1 register traffic, UART1 receive-line edges and receive interrupts, and the GIC's nIRQ; and it stops on a 300 ms watchdog. ROM images are generated from scratch copies of the test directories with the same tooling as the board images. The test is slow to simulate — about 11 s of CPU time per millisecond of SoC time, 3739 s for the 327 ms of the fourth image — because its delay loops are tens of milliseconds long; it was still cheaper than another board iteration each time, and it was the only way to see the fifth defect, which no board run could reach.

23.4 Result

Image	Change	Board, build 57, from the physical flash	VCS, FPGA RTL
v2	Loopback test re-enabled; handlers fixed to compile	HANG after 'UART0 TX --> UART1 RX' (interrupt flood)	—
v3	Receiver and interrupt off before printing; bounded handlers; console divider restored	HANG in 'UART1 TX --> UART0 RX' (WFI race)	PASS to the overrun tests
v4	Bounded poll in place of WFI	Both loopbacks PASS; HANG in RX overrun (clear/re-check race, error printed with TX off)	RX overrun PASS by timing; TX overrun FAIL (two writes never overrun)
v5	RX overrun waits for the line to go quiet and restores the console before printing; TX overrun writes three bytes; drain before every UART0 reconfiguration; SWRESET skipped on FPGA	TEST PASSED, 0 integration issues (27 Oct 12:45)	TEST PASSED at 101 ms

The upstream change (PR #6, Jira MBT-59) is confined to the test source; the SWRESET sub-test stays enabled by default. Two bench notes for the record: the runner's per-test summary line is cut at 140 characters, which hid the end of the third hang's last line for a while, and the NUL and garbage tail in those logs was the console adapter being switched to 2 Mbaud by the --baud-after option three seconds after boot, after the board had already hung — not board output.

24. Six pull requests and the Jira position (28 October)

Update, Revision 8: Daniel Newbrook merged all six pull requests (#5–#10) on 28 October at 09:20, before this revision's predecessor was issued; the 'open for review' statements below are historical. MBT-58 and MBT-59 can therefore go straight to Done (Section 32.3).

All fix branches were based on MegaSoC-Tech main as merged on 22 October and pushed on 28 October, one pull request each, reviewer Daniel Newbrook. The commit message of the Winbond branch was amended before the push: it had claimed a bootloader.c change (checking the start-up's return value) that is not in the commit and does not exist upstream either. Each description states how far its fix has been verified; the last two columns below are that statement.

PR	Branch	Ticket	Change	Verification
#5	fix/dap-system-apb-offset	MBT-58	PADDRSYS carries the window offset; DBGROMADDR corrected; dap_tests checks the ROM table class (3 commits)	Build 57 from the physical flash; VCS
#6	fix/uart-tests-hw	MBT-59	uart_tests runs to completion: six defects (Section 23)	Build 57 from the physical flash; VCS
#7	fix/timer-intstatus-race	MBT-61	Timer stopped before INTCLEAR in test_timer_intstatus()	Builds 55 and 57 from the physical flash
#8	fix/spi-wait-busy	MBT-57	Bounded wait for Busy to rise in SPI_WAIT_BUSY()	Hang reproduced on the board with a RAM-resident probe (FFh); fix compiled, not yet run in a stock ROM on the board
#9	fix/undriven-ports	MBT-42	PRDATA_SPI[31:16] tied off; RDMEMATTR/WRMEMATTR connected	In every FPGA build since October; no functional change
#10	fix/spi-startup-winbond	MBT-47	Winbond W25Qxx branch in SPI_STARTUP()	Sequence validated on the W25Q32FW through the monitor ROM (10 of 10 tests boot with it); the library code itself was compiled only into build 56's ROM, whose console rate was wrong — unobserved

Jira (project MBT) on 28 October. Atlassian has been reachable from the server again since 26 October; the paste-ready log of Revision 6 is now a record, not a queue.

Ticket	State	Position
MBT-42, 47, 57	Done	Closed on 21 October when the fixes were applied in the FPGA tree; the upstream changes are PRs #9, #10 and #8.
MBT-48, 55	In Review	Fixed on the AHB-QSPI master; Done once megasoc_tech's submodule pointer is bumped (row 77).
MBT-50, 56	Done	PR #4 merged 22 October.
MBT-58	To Do	Fixed and verified on build 57; proposed In Review with PR #5, awaiting the go.
MBT-59	To Do	Fixed and verified on build 57; proposed In Review with PR #6, awaiting the go.
MBT-43	Open	Proposed close: CLK1HZ is now constrained in haps_top_clocks_impl.xdc.
MBT-44	Open	Proposed obsolete: timing has been met since build 42.
MBT-61, 62	To Do (new, 28 Oct)	Timer test race (fix PR #7) and the c_print_str 79-character limit (fix not yet submitted). Bug, Low.
MBT-5 … 22	various	Shiva's tickets (SDIO, GIC-400, DRAM_AXI, EXP_M, PL011 RT, QSPI IRQ); untouched.

25. Current status (28 October 2026) — supersedes Sections 11 and 19

Area	Status	Evidence
CPU first fetch, ROM, SRAM, boot console	DONE, on hardware	Banner every boot since build 42.
UART0 both directions; UART1 and PL011 loopbacks	DONE, on hardware	pl011_uart_tests and the corrected uart_tests PASS from the flash (build 57).
Interrupts through the GIC-400; PPU power sequencing	DONE, on hardware	gic_tests and low_power_tests PASS.
QSPI controller, flash cache, XIP window; programming and booting the physical flash	DONE, on hardware	Every test image boots from the W25Q32FW with no SRAM preload (builds 55–57).
CPU caches; stock boot sequence through the monitor ROM	DONE, on hardware	Build 54 onward (Section 18.4).
Stock SoC Labs bootloader ROM with the Winbond library code	OPEN	Build 56's ROM console rate was wrong (row 76); rebuild with the corrected ROM to prove PR #10 end to end.
System access to the CoreSight debug block	DONE, on hardware	dap_tests PASS on build 57 (Section 22). ROM table configuration and window size open for SoC Labs (row 71).
SoC Labs test suite	10 of 10 buildable tests PASS	Build 57, 27 October, all from the physical flash (Section 25.1). sdio_tests needs a card; coremark and dma350_tests do not build upstream.
Timing	DONE	Build 57: WNS +14.240 ns, WHS +0.009 ns, all constraints met, one clock domain.
Upstream	PRs #7 and #4 merged; #5–#10 open	Section 24. Submodule pointer bumps outstanding (row 77).
Jira	Tickets to MBT-62	MBT-58/59 await In Review; 43/44 to close; 61/62 new (Section 24).
Reset controller in the FPGA snapshot	OPEN	Row 75; the SWRESET sub-test is skipped on FPGA builds.
SoC Labs VCS flow, flash boot	OPEN	Row 78.

25.1 Test matrix, build 57 (27 October) — every image from the physical on-board flash, no SRAM preload

Test	Result	Note
hello_world	PASS	Both caches on.
mem_tests	PASS	
timer_tests	PASS	Corrected image (MBT-61); the unmodified test reports 1 error from flash and passes from RAM.
neon_tests	PASS	
peripheral_test	PASS	
gic_tests	PASS	
low_power_tests	PASS	
pl011_uart_tests	PASS	Loopback under FPGA_BUILD.
dap_tests	PASS	First pass ever; MBT-58 fix and the CIDR check (Section 22).

uart_tests	PASS	Corrected test, fifth image (MBT-59); SWRESET sub-test skipped on FPGA builds (row 75).
sdio_tests	N/A	No card fitted.
coremark, dma350_tests	—	Do not build upstream.

25.2 Timing progression, builds 54–57

Build	WNS (ns)	WHS (ns)	Note
54	+14.072	+0.010	L2 tag integration (Section 18)
55	+15.130	+0.009	Same sources rebuilt from the GitHub work folder
56	+14.240	+0.009	Stock bootloader ROM + cpu_ss debug-window fix
57	+14.240	+0.009	Monitor ROM v5.2 + cpu_ss debug-window fix; 10 of 10 tests

26. Corrections to Revision 6, revert register additions, and recommended next steps

26.1 Positions of Revision 6 corrected by this revision

Revision 6 position	Where	Corrected position
dap_tests fails most likely because the debug power domain is never powered without a debugger.	Row 63; Sections 19 and 20.1 item 4	Address decode: the DAP-Lite system APB port received the full system address (row 69). Every debug power request was granted.
uart_tests has four defects: NULL receive buffer, console transmitter disabled, framing, 2 Mbaud console.	Row 65; Section 17.5	Six defects (row 73). The 'NULL buffer' is two local pointers shadowing the static buffers; the console and framing items are real but the smaller part; three races only show with code fetched from flash.
Divider 10 puts a CMSDK UART at 2 Mbaud.	Section 17.5; row 65; the runner's --baud-after option	1.25 Mbaud: the divider's minimum is 16, so 10 gives PCLK/16. Neither rate is readable by the bench adapter; the corrected test never changes the console rate.
Two QSPI controller fixes to send upstream.	Section 20.1 item 1; register item 36	Already fixed on the AHB-QSPI master; the action is a submodule pointer bump in megasoc_tech (row 77), not a code change.
Atlassian unreachable; Jira updates paste-ready.	Section 19; Section 20.1 item 2	Reachable again from 26 October; MBT-61 and MBT-62 posted 28 October; the remaining transitions await the go.

26.2 Tape-out revert register additions

Items 1–41 (Sections 12 and 20) still apply with the changes noted in Section 26.1. New items from this period:

#	Item	Type	Action before tape-out
42	megasoc_cpu_ss.sv: PADDRSYS carries the offset within the DEBUG window	RTL, shared with the ASIC	Must be in the ASIC netlist (PR #5); without it no system access to the CoreSight ROM table works.
43	megasoc_cpu_ss.sv: DBGROMADDR = 28'h0001300	RTL, shared with the ASIC	Must be in the ASIC netlist (PR #5).
44	DapRomDefs.v unconfigured; A53 ROM table shadowed; 32 KB DEBUG window reaches the ROM table only	ASIC integration decision	Configure the table and decide the window before tape-out; the FPGA runs with the unconfigured table and the test accepts it.
45	-DSKIP_SWRESET_TEST for FPGA test builds; scratch directories uart_tests_hw and uart_tests_rom; local dap_tests.c edit	FPGA / test only	Drop when PRs #5 and #6 merge; never define SKIP_SWRESET_TEST for silicon builds.
46	megasoc_reset_ctrl absent from the FPGA fork	FPGA-only gap	Port the upstream controller into the FPGA tree so the software reset path is exercised on hardware; no ASIC action.
47	megasoc_tech submodule pointers: sl_ahb_qspi (a16f13e, buggy) and sdspi (315fe083, not on GitHub)	Integration	Bump both before the ASIC netlist is taken.
48	Board ROMs built from the board tree only (uart_stdout for 20 MHz / 115200 lives on local/haps-fpga)	Process	Rebuild the stock bootloader ROM with the corrected divider and prove the deliverable boot path once.

26.3 Recommended next steps

1. Get PRs #5–#10 reviewed and merged; then pull megasoc_tech, drop the local dap_tests.c edit and the scratch test directories, and rebase local/haps-fpga onto the new main.

2. Jira: MBT-58 and MBT-59 to In Review with the PR links, Done when merged; close MBT-43; MBT-44 as obsolete; MBT-48 and MBT-55 Done after the pointer bump; comments on MBT-61 and MBT-62 when their fixes merge.

3. Bump the AHB-QSPI pointer in megasoc_tech to the fixed master and resolve the sdspi pointer. This is what carries the controller findings of Section 17.2 into the ASIC netlist.

4. Debug block: configure DapRomDefs.v and decide the DEBUG window — reach CPU0's debug registers at offset 0x10000, or document that system access is to the ROM table only.

5. Port megasoc_reset_ctrl into the FPGA snapshot and run uart_tests without the skip define; this also exercises the software reset path on hardware for the first time.

6. Rebuild the stock bootloader ROM with the corrected console divider into a bitstream and boot it once, to prove the deliverable boot path and PR #10's library code on the board; then decide which ROM a deliverable FPGA build carries.

7. Submit the one-line c_print_str fix (MBT-62).

8. Fix the SoC Labs VCS flow's flash-jump abort so flash-booted tests can be verified upstream; the FPGA-design testbenches (n25/sim/dap and uart) are the template.

9. Keep the RTL simulation of the FPGA design as the first tool for any anomaly: it found the cache stop, the debug-window bug and the TX-overrun defect, each within hours of being pointed at the symptom.

10. Low priority: a card for sdio_tests; remove the auto-reset hold-off and the logger taps from any deliverable FPGA build.

27. DDR4 step 1 — the Xilinx DDR4 IP alone on the SODIMM (28 October)

27.1 Task and approach

Jira MGSOC-34, "FPGA Verify DRAM integration (Can just use xilinx MIG)", a sub-task of MGSOC-18 "FPGA Implementation" created on 18 June and assigned to the author with no description. The ASIC's DRAM path is the Synopsys UMCTL2 controller with the LPDDR4 multiPHY (hard TSMC16 IP, repository Synopsys-LPDDR4-Integration-Subsystem, submodule megasoc_tech/logical/lpddr4_subsystem) and cannot be synthesised for the FPGA; no FPGA DRAM wrapper existed on GitHub. The FPGA path therefore uses the Xilinx DDR4 IP (ddr4 v2.2 in Vivado 2024.1) on the HAPS-SX's SODIMM, configured exactly as the vendor's proven HAPS-SX design (SoC-Labs/HAPS-ZynqMP-Daughter-Board, docs/reference/ddr4-configuration.md): Micron MTA16ATF2G64HZ-2G3, 16 GB x64 non-ECC, two ranks, DDR4-2133 (tCK 938 ps), 4:1 controller ratio, 299.85 MHz differential reference on AF51/AF52, CL 15 / CWL 11, data mask without DBI, ROW_COLUMN_BANK mapping, AXI 512-bit / 34-bit address / ID width 1 at 266.5 MHz. One deliberate departure: AXI narrow bursts are enabled — the vendor had them off and stalled on sub-512-bit writes.

Pins: the 123 DDR4 lines were generated from SoC-Labs-HAPS-SX-Tools iomap_csv_rev0b/pinfile.csv (DDR4 section). They sit in banks 29, 30 and 31, which the device query mapped to clock regions X0Y10, X0Y11 and X0Y12 — all in SLR2, the left I/O column — with the reference clock on a global-clock pair in bank 30; POD12 / SSTL12 DCI with the DCI cascade from bank 30. The board has no PAR or ALERT lines, so parity is off. A byte-lane check before the build placed every DQ byte with its DQS pair and DM in one lane, the strobes on DBC/QBC pins and all 33 address and control lines in bank 30. The older haps-sx_vu19p.def lists ACT_N on CA49, which is the flash clock pin: wrong, the CSV's AE48 was used.

Placement context from build 57's reports: the SoC occupied SLR0 (31 % of CLBs, 67 % of block RAM) and SLR1 (7 %, 57 %), with X0–X1 Y10–Y12 empty, and only 79 of 23,040 SLR1–SLR2 super-long lines in use. The plan was therefore to keep the IP and the AXI converters in SLR2 next to the DDR4 banks and to cross the SLR boundary on the slow 20 MHz side.

27.2 Example-design build (11:12–11:53)

gen_ddr4_ip.tcl creates the IP and opens its example project (traffic generator SIMPLE_TG); haps_ddr4_ex_top.sv wraps it (USER_PB2 inverted into the IP's active-high sys_rst; LED_G1 = calibration complete; LED_R0 = traffic-generator compare error) and the generated example_design.xdc is disabled because its placeholder pins land on the SoC's clock and reset pins. Lessons for the IP's Tcl: set TimePeriod before MemoryType and MemoryPart; list_property_value

returns nothing for the dynamic enumerations; the clock type is CONFIG.System_Clock; there is no reset-polarity option.

Item	Result
Bitstream	haps_ddr4_ex_top.bit, md5 7eb74a77..., archived in bitstreams/ddr4_example_20261028/ with reports and sources
Timing	All constraints met: WNS +0.435 ns, WHS +0.010 ns. c0_sys_clk 299.85 MHz, MMCM 16/3/6 → memory clock 1066 MHz, user clock 266.5 MHz; the debug hub on the JTAG clock. 0 critical warnings
DRC	4 warnings, none on pins or the memory interface (3 × PDCN-1569 LUT equation on the debug hub's BSCAN switch, 1 × RTSTAT-10 unloaded IP-internal nets)
Placement	Everything in SLR2: X0Y10 15.3k flops, X0Y11 15.6k flops with the MMCM and 5 BUFGCE, X0Y12 2.3k; 23.9k LUTs, 33.5k FFs, 25.5 block RAM tiles, 3 DSPs. Nothing in the SoC's regions except the BSCAN flops

27.3 Bench, over JTAG from the server (12:15–12:55)

The HAPS-SX JTAG chain is served by hw_server at 127.0.0.1:3310 on srv03335 (target */xilinx_tcf/Digilent/HAPSSXB87822); check_ddr4_cal.tcl connects there, programs a bitstream and prints the IP's calibration status through the debug hub — no bench access needed. A plain connect_hw_server starts a useless local hw_server on 3121 instead and must be killed.

v1: CALIBRATION_FAIL.STATUS = FALSE, "No errors detected during calibration", every stage PASS or SKIP (DBI and VREF stages skipped by configuration, rank-1 repeats of per-bit steps skipped as designed), both ranks trained. Margins (rank 0, taps left/right): read window rising 36–45 / 37–45 and falling 34–40 / 35–41 per byte; write DQS-to-DQ window 0x37–0x3d / 0x36–0x3d; rank 1 similar — symmetric and uniform across the eight bytes, so the strobes are centred and the board routing is clean. v1 could not show the traffic generator's compare-error flag (LED_R0 only), so v2 added a VIO (timing met, WNS +0.369): after a reset pulsed through the VIO the IP re-calibrated, calib_s = 1 and err_s = 0 across three one-second samples with the heartbeat toggling. Step 1 complete at 12:55: the SODIMM works end to end with the Xilinx IP on this board, and the SoC integration needs no floorplanning beyond keeping the converters in SLR2.

28. DDR4 step 2 — the IP on the NIC-400 DRAM port (build 58, 28 October)

28.1 Design

haps_ddr4_bridge.sv replaces the 512 KB axi4_bram_slave on the DRAM port: DRAM_AXI (64-bit data, ID width 9, 33-bit address, SYS_CLK 20 MHz) → axi_cc_dram, the Xilinx AXI clock converter v2.1 with asynchronous FIFOs (20 MHz ↔ ui_clk 266.5 MHz) → axi_dw_dram, the AXI data width converter v2.1 (64 → 512, single thread, IDs stored in order because the 512-bit side has no ID) → ddr4_0 in the step-1 configuration (AXI 512-bit / 34-bit / ID 1, tied 0). The NIC-400 passes the low 33 address bits; the bridge ties the IP's bit 33 (the rank bit) low, so rank 0 — the first 8 GB of the 16 GB DIMM — is reachable: CPU 0x8000_0000–0x1_FFFF_FFFF maps to DRAM 2–8 GB and CPU

0x2_0000_0000–0x2_7FFF_FFFF to DRAM 0–2 GB. The firmware's DRAM_BASE 0x8000_0000 is unchanged; the library page table maps 2–4 GB Normal cacheable and everything above 4 GB Device, so the whole window is reachable (the upper part uncached).

Resets: the IP's sys_rst is PB1 only — the 10 s auto-reset and the monitor's 'g' never touch the IP, so DRAM contents survive both. The converters and the IP's AXI shim are held in reset while the SoC is in reset, calibration is incomplete or ui_clk_sync_rst is asserted, released through four-stage synchronisers in each domain; a DRAM access before calibration completes stalls the CPU (AWREADY/ARREADY low) rather than returning an error. localparam USE_DDR4 in megasoc_tech_wrapper.sv selects the bridge (1) or the block RAM slave (0) under a generate; new ports DDR4_SYS_CLK_P/N, DDR4_SYS_RST and DDR4_CALIB_COMPLETE (LED_G1 on BE14) are threaded through megasoc_chip.v, megasoc_system.sv and haps_top.sv, and DDR4_CK_T/C widened to two clock pairs. haps_top_ddr4_impl.xdc (implementation only) carries the 123 pins renamed to the top-level ports, the DCI cascade, the LED and set_clock_groups -asynchronous between HAPS_CLK (with CLK1HZ) and the IP's reference clock and its MMCM/PLL children.

28.2 Bridge simulation (xsim, 28 October)

tb_bridge.sv drives the bridge with the real converter IP models and ddr4_0_sim_model.sv, a behavioural stand-in for the DDR4 IP (ui_clk, reset and calibration timing, 512-bit AXI slave over a sparse memory, stalling handshakes). 203 checks, 0 errors: a write issued before calibration completes only after it; INCR and WRAP (A53 line-fill) bursts; byte, halfword and word writes and narrow reads; addresses above 4 GB; 0x2_0000_0040 lands on DDR4 line 1 (the alias); responses in order with the right IDs; a SoC reset while idle followed by traffic. Re-run with ddr4/sim/mk_brsim.tcl.

28.3 Build 58 (15:16–17:58, three attempts)

Attempt 1 stopped at the elaboration gate because the out-of-context IP netlists did not exist yet — the script now runs the IP synthesis first (synth_design -rtl needs the OOC checkpoints or reports 'module not found'). Attempt 2 failed in synthesis with seven 'Synth 8-2914 Unsupported RAM template' errors attributed to sdspi/rtl/sdax_s2mm.v line 63 — a bogus attribution (the last parsed file; build 57's RAM messages had pointed at qspi_flash_emu.sv the same way). The real cause, reproduced standalone: with the bridge on the DRAM port the flash emulator's backdoor ports are tied to constants and Vivado can no longer map seven of its eight lane RAMs. The emulator is now instantiated only when USE_FLASH_EMU = 1 (generate guard, outputs tied otherwise). Side effect worth recording: builds 50–57 had kept the dead emulator's 512 KB of block RAM alive through that backdoor read path; build 58 frees it. Attempt 3 is the bitstream.

Item	Build 58 result
Bitstream	haps_top_build58_ddr4_20261028.bit, md5 5af9c318…; pre-change copies *.pre_build58 beside each edited file; build script rebuild_bitstream58.tcl (creates the three IPs when absent, OOC synthesis, elaboration gate, synth_1, constraint guard, impl_1)
Timing	Met: WNS +0.510 ns, WHS +0.010 ns, WPWS +0.065 ns, 0 failing of 336,969 endpoints. HAPS_CLK WNS 29.77 ns; the 0.51 ns worst path is inside the IP's MMCM/PLL domains as in step 1. HAPS_CLK ↔ ui_clk reported as asynchronous groups (the reference clock is created on the top-level port and is named DDR4_SYS_CLK_P, not c0_sys_clk_p; the constraint names both)
DRC	0 errors, 20 warnings = build 57's 17 (DSP pipelining in the A53, one gated clock in the peripheral subsystem, RTSTAT-10) + the 3 PDCN-1569 LUT-equation warnings of the debug hub; 0 latches
Placement	The placer moved the WHOLE SoC into SLR2 beside the DDR4 banks (SLR2: 40.6 % CLBs, 270.5k LUTs, 114.5k flops; SLR0/1/3 under 300 LUTs each; 180 SLLs SLR2–SLR3, 24 SLR2–SLR1) instead of SLR0/1 as in build 57. No floorplan constraint is active; timing met, left as is. Bridge: 19,953 LUTs, 18,204 flops, 25 block RAM tiles
Board (JTAG, 17:53)	The DDR4 IP calibrates inside the SoC design: CALIBRATION_FAIL.STATUS FALSE, no errors, RANK0 stages 01–27 PASS or SKIP exactly as in step 1 (cal_status_build58_20261028.txt); re-confirmed on 29 October after the board was re-programmed

28.4 Test program ddr4_fpga_tests

A new bare-metal test (megasoc_tech/software/src/ddr4_fpga_tests, Arm Compiler 6.22, two link variants: the dram scatter file to execute from DDR4 at 0x8000_0000, the default scatter file to execute from the flash XIP window with data in the on-chip SRAM) targets the DRAM window directly. Version 1 had four tests; by version 8 (30 October) it has nine, listed in Section 29.1 with the reason each was added. Every version prints a per-test result line and ends with "** TEST PASSED **" and the suite's end marker; from version 3 each test also prints its duration in generic-counter ticks (20 MHz).

29. Bench, 29 October: the DRAM path works, and a second N25 instance (build 58)

29.1 Eight versions of the test in one day

Version	Change	Result on build 58
v1 (28 Oct)	T1 64 KB patterns; T2 one cell per address bit 3..32 at DRAM_BASE + 2^k plus the 0x2_0000_0000 alias and the top of the window; T3 4 MB hashed block through the caches; T4 8/16/32-bit stores to Device memory above 4 GB	RAM: 'running from DDR4', then HANG — T2's cells for k = 3..16 lie inside the running image (boot code, vectors at +0x800, library, RW data) and overwrote it. Flash: T1 passed, T2 cell k = 31 → data abort ESR_EL3 0x96000044, FAR 0x1_0000_0000 (row 82)
v2 (10:16)	mmu_extend() widens TCR_EL3 to 39-bit VA / 40-bit PA before any access; T2 cells moved to +2 MB + 2^k	RAM: PASS 7.7 s, T1–T4 0 errors — the DRAM port with the Xilinx IP is functionally proven from the CPU. Flash: T1 and T2 pass, no end marker within the 60 s suite timeout during T3
v3 (10:32)	Per-test CNTPCT times and T3 progress lines; suite --timeout 200	Flash: T1 330.8 M ticks (16.5 s for 64 KB), T2 0 errors, T3 pass 0 1.67 G ticks — slowness, not a hang: uncached XIP code fetch (row 83); counter = 20 MHz
v4 / v5 (11:26)	The test reads SCTLR_EL3 and CPUECTLR and enables the caches itself	SCTLR_EL3 0x40c52839 (C=0, I=0) in BOTH modes before the test, CPUECTLR 0 (row 80). With C=I=1 flash mode is as fast as DDR4 (T1 786 K vs 771 K ticks). T1, T2, T4 0 errors; T3 FAILS in both modes: 291,801 (RAM) / 708,736 (flash) of 1,048,576 words — every wrong word is another line of the SAME L2 set
v6 (11:53)	SMPEN set before enable_caches() (row 81); T5 cache-resident L2-set aliasing (16 and 18 lines of one set, read while resident and again from DRAM after clean+invalidate); T6 1 MB written cached and read with the data cache off, then written uncached and read cached	T5 0 errors — the tag and way handling is right and SMPEN was not the cause. T3 still fails (34,864 / 32,360). T6: cached writes read back uncached 73,136 / 83,680 of 131,072 words WRONG; uncached writes read back cached 0 errors — read bursts and single writes are right, the data the L2 writes back is wrong
v7 (12:12)	T8: 64 KB of dirty DDR4 lines evicted by a 512 KB read sweep (write-backs while only reads are issued); T7: the same with the on-chip SRAM remapped Normal cacheable (RAM image only; the DRAM port then carries reads only)	T8: 7,536 of 8,192 words wrong (92 %), and the stale data came from a line that had only ever been written uncached — data the bridge never carried as a write. The DDR4 bridge is cleared. T7: SError ESR_EL3 0xBF000000 immediately after the page-table change (row 85)
v8 (30 Oct)	T7 cells moved from 0x81_0000 to 0x80_8000–0x80_FFFF, inside the 64 KB the NIC-400 decodes	Build 59: all nine tests 0 errors in RAM mode (Section 30)

29.2 Firmware facts learnt on the way

None of these is a DRAM problem; all are recorded as bug-register rows 80–83 with proposals for SoC Labs. (1) The monitor ROM leaves the CPU caches off and the library's boot.s enables the MMU only, so every RAM-mode run since build 47 — and every flash-booted test since build 51 — executed with C = I = 0; the DRAM port had only ever seen single 64-bit transfers. (2) enable_caches() never sets CPUECTLR.SMPEN. (3) TCR_EL3 = 0x00102F20 limits EL3 to 4 GB of VA and a 32-bit PA. (4) The monitor's Winbond 'n' boot jumps without the CG092 flash-cache enable, so flash-booted code runs about 30 times slower than the same code from DDR4 or through 'g'. (5) The generic counter runs at 20 MHz with CNTFRQ_EL0 unset. (6) A test that writes cells at DRAM_BASE + 2^k overwrites its own image; the cells now start at +2 MB.

29.3 The failure signature, and where it pointed

Every wrong word of versions 5–7 was traced (a Python reconstruction of the test's pattern functions over the logged addresses and values) to the pattern value of ANOTHER LINE OF THE SAME L2 SET: the exclusive-or of the expected and the actual source address always had bits 13:6 equal to zero (observed differences 0x338000, 0x27bc000, 0x24d4000, 0x828000, 0x24f4000, 0x2404000, 0x508000 — tag bits only). Three facts then separate the L2 from the bridge: T5 shows 16 and 18 lines of one set read back correctly while resident and again from DRAM, so tags and ways are indexed correctly; T6 shows cache-line fills (uncached writes read back cached) correct and dirty write-backs wrong; T8 shows a write-back carrying data the bridge had never been given as a write. The L2 was writing back another way's data from its DATA RAMs. The synthesis log named the instance: haps_top/g_l2_rams.u_ca53_l2_datarams/u_l2_dataram_0_low/u/mem_reg — Arm's TSMC 1 MB example wrapper (A[13:0] unmasked), present in the Vivado fileset beside the generic one and chosen by the tool as the active definition. Build 54's N25 fix (Section 18) had replaced the TAG RAM integration only; the data RAMs had been wrong since the project began and nothing had exercised them with evictions of two same-set lines while the caches were on.

29.4 Build 59 (16:31–19:17): the generic, index-masked wrappers for both RAM groups

RTL identical to build 58; only the fileset changed (rebuild_bitstream59.tcl, project file backed up as haps_top.xpr.pre_build59): disabled models/rams/tsmc_cln28hpm_L1_32I_32D_L2_1M/ca53_l2_datarams.v (the example build 58 synthesised), mem_integrations/FPGA/ca53_l2_datarams.v (SoC Labs' FPGA version — also wrong, row 84, never used) and the mem_integrations/CortexA53 duplicates of both wrappers; enabled, as the only definitions, Arm's generic ca53_l2_datarams.v and ca53_l2_tagrams.v over new/ca53_generic_ram_fpga.v (build 54). The script asserts that the compile order holds exactly one data and one tag wrapper before synthesis. Elaboration: 8 generic data RAM instances, 17 generic tag RAM instances, no TSMC macro under the L2 (the 33 remaining macro models are the L1 and TLB RAMs). Synthesis: 310 block RAM primitives against 522 — the constant l2_size mask lets Vivado trim the generic RAMs to 4096 × 64 per bank, so the fix saves block RAM rather than costing it. Timing met: WNS +0.448 ns, WHS +0.010 ns, WPWS +0.065 ns, 0 failing endpoints; DRC 20 warnings as build 58; 0 latches. Programmed over JTAG at 19:20: the DDR4 IP calibrates with no errors. The board has held build 59 since.

30. Bench, 30 October: build 59 verified, the SRAM SError explained, both regressions

30.1 ddr4_fpga_tests on build 59

Run	Result
06:05 flash mode, v7 (suite --winbond)	PASS, 3.1 s, "DDR4 total errors: 0" with the caches on. T8 — 7,536 errors on build 58 — now 0; T3 and T6 likewise. The L2 data RAM integration was the defect; the DDR4 bridge and IP are clean under allocation-driven evictions
06:01 RAM mode, v7	Ran to T7 and took the SError again. Root cause found in the design sources within the hour: the NIC-400 decodes 0x800000–0x80FFFF only (nic400_asib_A53_decode: 44'h800000 ≤ addr ≤ 44'h80FFFF; Socrates memory_map.xml, RAM range 0x10000) and the test's cells were at 0x81_0000 (row 85). The SIE-300 SRAM controller was cleared: cfg_gate_resp is tied low in megasoc_tech_wrapper.sv, so it stalls when gated and never returns SLVERR
06:30 RAM mode, v8	PASS, 2.5 s, all nine tests 0 errors. T1 64 KB × 5 patterns 0.77 M ticks (39 ms); T2 33 address cells 0.17 M; T5a/T5b one-set aliasing 0.19 M each; T3 4 MB × 2 passes 37.8 M (1.89 s, 0.44 s per write-and-clean pass); T6 1 MB both directions 8.9 M; T8 64 KB of dirty DDR4 lines evicted 0.55 M; T7 32 KB of dirty SRAM lines at 0x80_8000 evicted and verified 0.59 M; T4 narrow Device stores 0.13 M. SCTLR_EL3 found at 0x40c52839, SMPEN set, caches enabled → 0x40c5383d

30.2 Regressions on build 59

RAM mode from the SODIMM (06:33): hello_world, mem_tests, gic_tests, timer_tests, neon_tests, peripheral_test and pl011_uart_tests loaded into DDR4 by the monitor and executed from there — 7 of 7 PASS, the first time the SoC Labs tests have run from DRAM rather than from a block RAM. These images run with the caches off, as the library leaves them (row 80). Flash (07:09–08:05, build 57's images from the physical W25Q32FW): dap_tests, hello_world, mem_tests, neon_tests, peripheral_test, gic_tests, low_power_tests, pl011_uart_tests, timer_tests — 9 of 9 PASS. uart_tests_hw first reported HANG at 08:04; that was the host, not the board: the run had been given --baud-after, which switches the console adapter to 2 Mbaud three seconds after the boot line, while the corrected test (v5) keeps the console at 115200 throughout (Section 26.1, third row), so the tail of its output arrived as garbage. Re-run without the option at 10:37: PASS, "Total integration issues in UART is 0". Flash regression 10 of 10, the same set as build 57.

30.3 What the DDR4 evidence covers

From the CPU, with the caches on and off, in both boot modes: instruction fetch and data from the SODIMM (the images and the monitor's load path); single 64-bit transfers; 8-, 16- and 32-bit stores with their strobes, including above 4 GB; 33 address bits one at a time, the 0x2_0000_0000 alias of DRAM offset 0 and the last word of the 8 GB window; cache-line fills (WRAP bursts) by the megabyte; explicit clean+invalidate write-backs; allocation-driven dirty evictions with reads in flight; the same evictions towards the on-chip SRAM. Not covered: rank 1 and the second 8 GB

(unreachable by design, address bit 33 tied low); sustained bandwidth (the SoC side runs at 20 MHz, so the IP is never stressed); ECC (the SODIMM is non-ECC).

30.4 Pull requests and the Jira position (30 October)

MegaSoC-Tech: Daniel Newbrook merged PRs #5–#10 on 28 October at 09:20 (MBT-58 dap fix, MBT-59 uart_tests, MBT-61 timer race, MBT-57 wait-busy bound, MBT-42 undriven ports, MBT-47 Winbond start-up); MegaSoC-Tech main now carries every fix of Sections 22–24. MegaSoC-Project: the DDR4 work is contributed as PR #8 on the fpga/targets/haps_sx/bringup folder — the bridge and the updated top level, the DDR4 constraints, the step-1 scripts and the bridge testbench, the build scripts of builds 58 and 59, the regenerated megasoc_tech_wrapper.sv patch and the build-58 diffs of megasoc_chip.v and megasoc_system.sv, the test program with its images and the seven RAM-mode regression images, the latest host tool, the notes of builds 55–59, the updated bug log, this record, and a README that corrects its predecessor's statement that the FPGA build uses the TSMC example L2 data RAM wrapper. Nothing Arm-licensed is in it; the FPGA work area stays on the server. Jira: nothing was posted between 28 and 30 October; the running log holds the paste-ready text for MGSOC-34 (comment and transition) and for the new tickets of Section 32.3, awaiting the go.

31. Current status (30 October 2026) — supersedes Sections 11, 19 and 25

Area	Status	Evidence
CPU first fetch, ROM, SRAM, boot console	DONE, on hardware	Banner every boot since build 42.
UART0 both directions; UART1 and PL011 loopbacks	DONE, on hardware	pl011_uart_tests and the corrected uart_tests PASS from the flash (builds 57 and 59).
Interrupts through the GIC-400; PPU power sequencing	DONE, on hardware	gic_tests and low_power_tests PASS.
QSPI controller, flash cache, XIP window; programming and booting the physical flash	DONE, on hardware	Every test image boots from the W25Q32FW with no SRAM preload (builds 55–59).
CPU caches — tag AND data RAM integration	DONE, on hardware	Build 59: 4 MB block, 1 MB cached/uncached cross-checks, dirty evictions to DDR4 and to SRAM, all 0 errors (Sections 29–30). Builds 54–58 had the data RAMs wrong (row 79).
DDR4 SODIMM on the NIC-400 DRAM port (MGSOC-34)	DONE, on hardware	Step 1 and step 2 verified 28–30 October: calibration, 203-check bridge simulation, ddr4_fpga_tests v8 from DDR4 and from flash, 7 SoC Labs tests executing from the SODIMM (Sections 27–30).
System access to the CoreSight debug block	DONE, on hardware	dap_tests PASS on builds 57 and 59. ROM table configuration and window size open for SoC Labs (row 71).
Stock SoC Labs bootloader ROM with the Winbond library code	OPEN	Build 56's ROM console rate was wrong (row 76); rebuild with the corrected ROM to prove PR #10 end to end.

SoC Labs test suite	10 of 10 buildable tests PASS from flash; 7 of 7 RAM-mode tests PASS from DDR4	Build 59, 30 October (Section 31.1). sdio_tests needs a card; coremark and dma350_tests do not build upstream.
SoC Labs tests with the caches on	OPEN (library)	The tests run with C = I = 0 unless the ROM enabled the caches (row 80); ddr4_fpga_tests is the only image that runs cached. Proposal in Section 32.3.
Timing	DONE	Build 59: WNS +0.448 ns, WHS +0.010 ns, all constraints met; HAPS_CLK and the DDR4 user clock in asynchronous groups.
Upstream	MegaSoC-Tech PRs #4–#10 merged; MegaSoC-Project PR #7 merged, PR #8 (DDR4, this record) open	Sections 24 and 30.4. Submodule pointer bumps outstanding (row 77).
Jira	Tickets to MBT-62; MGSOC-34 still To Do	Proposals of Section 32.3 awaiting the go: MGSOC-34 comment and transition, MBT-58/59 Done, new tickets for rows 79–85.
Reset controller in the FPGA snapshot	OPEN	Row 75; the SWRESET sub-test is skipped on FPGA builds.
SoC Labs VCS flow, flash boot	OPEN	Row 78.

31.1 Test matrix, build 59 (30 October)

Test	From the physical flash	From DDR4 (RAM mode)	Note
ddr4_fpga_tests	PASS (v7)	PASS (v8)	Nine tests, caches on; the only cached image. T7 (SRAM evictions) RAM mode only.
hello_world	PASS	PASS	
mem_tests	PASS	PASS	
timer_tests	PASS	PASS	Corrected image (MBT-61) from flash; stock image from DDR4.
neon_tests	PASS	PASS	
peripheral_test	PASS	PASS	
gic_tests	PASS	PASS	
low_power_tests	PASS	—	No RAM image built.
pl011_uart_tests	PASS	PASS	Loopback under FPGA_BUILD.
dap_tests	PASS	—	MBT-58 fix; the ROM table class check.
uart_tests	PASS	—	Corrected test v5 (MBT-59); run without --baud-after.
sdio_tests	N/A	N/A	No card fitted.
coremark, dma350_tests	—	—	Do not build upstream.

31.2 Timing progression, builds 54–59

Build	WNS (ns)	WHS (ns)	Note
54	+14.072	+0.010	L2 tag integration (Section 18)
55	+15.130	+0.009	Same sources rebuilt from the GitHub work folder
56	+14.240	+0.009	Stock bootloader ROM + cpu_ss debug-window fix
57	+14.240	+0.009	Monitor ROM v5.2 + cpu_ss debug-window fix; 10 of 10 tests
58	+0.510	+0.010	DDR4 IP on the DRAM port; the worst path is inside the IP's 266.5 MHz domain (HAPS_CLK slack 29.77 ns); whole SoC placed in SLR2
59	+0.448	+0.010	Generic L2 data RAM wrappers; 310 block RAM primitives instead of 522; board baseline

32. Corrections to Revision 7, revert register additions, and recommended next steps

32.1 Positions of Revision 7 corrected by this revision

Revision 7 position	Where	Corrected position
The cache-enable stop was root-caused to Arm's TSMC example L2 tag RAM integration and fixed in build 54; both CPU caches work.	Sections 18 and 19; row 67; Section 25 'CPU caches DONE'	Build 54 corrected the TAG RAMs only. The DATA RAMs kept the same unmasked example until build 59; both caches 'worked' on builds 54–58 only for working sets that never evicted dirty lines of two same-set addresses (row 79).
Seven of eleven tests pass from RAM (build 48); tests pass from the flash with both caches on (builds 51–57).	Sections 16, 17 and 25.1	The tests passed, but with the CPU caches off in every RAM-mode run and in every flash-booted test after the monitor's jump (row 80). Only hello_world through the stock boot sequence (build 54) and now ddr4_fpga_tests have run cached.
The FPGA build uses Arm's generic L2 tag wrapper together with the TSMC example ca53_caches_tlb_rams.v, ca53_l2_datarams.v and ca53scu_l1d_tagrams.v.	Upstream README of the bring-up contribution (PR #7)	The example ca53_l2_datarams.v is the defect of row 79; the generic data RAM wrapper is required too. Corrected in the README of PR #8.
Six fix pull requests (#5–#10) are open on MegaSoC-Tech for review.	Title table; Sections 24 and 25	Merged by Daniel Newbrook on 28 October at 09:20, before Revision 7 was issued.
A 'HANG' verdict from the suite means the board stopped.	Sections 23 and 25; haps_qspi.py	A HANG is also produced by code running uncached from flash (row 83, ~30× slower) and by a wrong --baud-after on the host (Section 30.2). The suite's --timeout and the per-test tick counts separate the three.

32.2 Tape-out revert register additions

Items 1–48 (Sections 12, 20 and 26) still apply with the changes noted in Section 32.1. New items from this period:

#	Item	Type	Action before tape-out
49	haps_ddr4_bridge.sv, the three Xilinx IPs (axi_cc_dram, axi_dw_dram, ddr4_0), USE_DDR4 = 1, the DDR4 ports through megasoc_chip.v / megasoc_system.sv / megasoc_tech_wrapper.sv, haps_top_ddr4_impl.xdc	FPGA only	Not for the ASIC: its DRAM path is the Synopsys UMCTL2 + LPDDR4 PHY. The port additions are guarded by USE_DDR4 and FPGA_BUILD.
50	L2 RAM integration: Arm generic ca53_l2_datarams.v and ca53_l2_tagrams.v over ca53_generic_ram_fpga.v (TSMC example and mem_integrations duplicates disabled)	FPGA fileset; ASIC check	The ASIC uses mem_integrations/CortexA53 over the TSMC16 macros. Confirm that integration masks or sizes the set index for L2_CACHE_SIZE = 256 KB — the example tree both integrations start from is for 1 MB (rows 67, 79).
51	ddr4_fpga_tests: sets CPUECTLR.SMPEN, enables the caches, widens TCR_EL3, remaps the on-chip SRAM cacheable (T7)	Test only	Nothing to revert; the library proposals of rows 80–82 are the durable form.
52	qspi_flash_emu instantiated only under USE_FLASH_EMU (generate guard)	FPGA only	No ASIC action; the emulator never existed there.
53	NIC-400 RAM region 64 KB at 0x80_0000 against a 128 KB FPGA SRAM wrapper	Integration check	Decide the SRAM size the SoC advertises; align the decode, the wrapper and sys_memory_map.h (row 85).

32.3 Recommended next steps

1. Jira, on the go: MGSOC-34 — comment with the Section 27–30 summary and move to In Review (or Done for step 2); MBT-58 and MBT-59 to Done (PRs #5 and #6 merged); new tickets: High, 'FPGA L2 data RAM integration uses the TSMC 1 MB example (unmasked set index) — evictions return another line's data' (row 79, fixed in build 59); Low, 'On-chip SRAM: NIC-400 decodes 64 KB, FPGA wrapper holds 128 KB, no size symbol in sys_memory_map.h' (row 85, tell Daniel first); Low, the library items of rows 80–82 (Arm-licensed files: proposals with the exact register values, no code); Low, monitor cache_on() before the Winbond jump (row 83); for the record, row 84.

2. Get MegaSoC-Project PR #8 reviewed and merged; then rebase local/haps-fpga of megasoc_tech onto the merged main (PRs #5–#10) and drop the local dap_tests.c edit and the scratch test directories.

3. ASIC: confirm the L2 data and tag RAM integration used for the TSMC16 netlist masks or sizes the set index for the 256 KB L2 (register item 50). This is the one finding of the period that could reach silicon.

4. Run the SoC Labs tests with the caches on: either enable them in the library start-up (with SMPEN) or in the monitor before the jump, then repeat the RAM-mode regression from DDR4 and the flash regression; also run lpddr4_dram_tests (generic DRAM tests at DRAM_BASE) on the FPGA DDR4 now that a DRAM exists.

5. Bump the AHB-QSPI pointer in megasoc_tech to the fixed master and resolve the sdspi pointer (row 77).

6. Debug block: configure DapRomDefs.v and decide the DEBUG window (row 71).

7. Port megasoc_reset_ctrl into the FPGA snapshot and run uart_tests without the skip define (row 75).

8. Rebuild the stock bootloader ROM with the corrected console divider and boot it once on build 59 (row 76); submit the one-line c_print_str fix (MBT-62).

9. Fix the SoC Labs VCS flow's flash-jump abort (row 78).

10. Low priority: a floorplan decision now that the placer puts the whole SoC in SLR2 (timing is met, so none is needed yet); a card for sdio_tests; remove the auto-reset hold-off and the logger taps from any deliverable FPGA build.

Appendix A — Environment reference

Item	Value
Development server	srv03335.soton.ac.uk (account ad1u25), Vivado 2024.1
Workstation	Windows, Vivado 2026.1 — retired as a build machine on 12 October 2026; its project copy (Desktop haps_top_project) is no longer synchronised.
FPGA working area (server)	/home/ad1u25/megasoc_project/FPGA_Tasos/haps_top/ — Vivado project haps_top.xpr (backups haps_top.xpr.pre_build59 at the project root and in bitstreams/build59_...); bitstreams/<build>_<date>/ holds every bitstream with reports, notes, firmware and build script; ddr4/ holds the DDR4 step-1 scripts, pin constraints, check_ddr4_cal.tcl and the bridge testbench (sim/).
Repository	github.com/SoC-Labs/MegaSoC-Project (public; submodules MegaSoC-Tech, AHB-QSPI, SoCTools-Flow, ASIC-Flow). The local work folder /home/ad1u25/megasoc_project tracks it since 22 October 2026; FPGA_Tasos/ is git-ignored (Arm IP). Upstream fix branches are prepared in a worktree of megasoc_tech.
Target	xcvu19p-fsva3824-2-e; 4 SLRs of 45 clock regions each (rows Y0–4, 5–9, 10–14, 15–19); 540 RAMBFIFO36 sites per SLR
Design clock	HAPS_CLK, 20 MHz (50.000 ns period)
Debug UART logger	debug_uart_probe.sv, PMOD3 IO[3] (BF19), 115200 8N1; the sole debug channel
Jira	soclabs.atlassian.net — projects MGSOC and MBT (bug tracker)
Bench PC	haps-dev: console /dev/ttyACM0 (UART0, 115200 8N1), logger /dev/ttyUSB0. The suite runner is ~/build56/haps_qspi.py (cmd, cmds, program, load, go, boot, suite [--flash|--winbond] [--timeout S], term); --baud-after only for a test that changes the console rate — NOT the corrected uart_tests (a false HANG otherwise). Images: ~/firmware/ (RAM-mode *_dram.bin and the ddr4_fpga_tests images), ~/flash_images/ (build 57's flash set). Suite summary lines are the last lines of output cut at 140 characters; the full capture is in ~/haps_captures/suite_<stamp>/<test>.log. Files are pulled from the server with scp from srv03335.soton.ac.uk.
Board	Programmed from srv03335 (ConfPro test_reset_timing.csprj, GCLK1 20 MHz; Vivado hardware manager, or ddr4/check_ddr4_cal.tcl <bit> which also prints the DDR4 calibration status). JTAG: hw_server 127.0.0.1:3310, target */xilinx_tcf/Digilent/HAPSSXB87822. On-board flash W25Q32FW on bank 21 (CA49 CLK, CB49 CS, CC49/CC50/CA51/CB51 D0–D3). DDR4: Micron MTA16ATF2G64HZ-2G3 SODIMM (16 GB, 2 ranks, rank 0 used) on banks 29/30/31 (SLR2), reference clock 299.85 MHz on AF51/AF52; LED_G1 (BE14) = calibration complete; IP reset = PB1 (HAPS_RESET_N), never the SoC auto-reset.
Backups	~/backup_megasoc_project_20261021/ (whole project before the GitHub switch, 20 GB); ~/Desktop/haps_top_backup_20261028/ (haps_top project at build 57, 1.5 GB tar.gz with md5, manifest and README); haps_top.xpr.pre_build59 (project file before the fileset change of build 59).

DDR4 IP	Xilinx ddr4 v2.2 (Vivado 2024.1) as the vendor's HAPS-SX reference: DDR4-2133 tCK 938 ps, 4:1, CL 15 / CWL 11, DM no DBI, ROW_COLUMN_BANK, AXI 512-bit / 34-bit / ID 1, narrow bursts enabled, RD_PRI_REG, MMCM 16/3/6 → memory clock 1066 MHz, user clock 266.5 MHz. Bridge: axi_cc_dram (AXI clock converter v2.1, asynchronous) + axi_dw_dram (AXI data width converter v2.1, 64→512). CPU window 0x8000_0000–0x2_7FFF_FFFF = DRAM rank 0 (2–8 GB then 0–2 GB).
Firmware build	source set_env.sh in /home/ad1u25/megasoc_project, then make -B COMPILE_GCC=0 AARCH64=1 in megasoc_tech/software/src/<test>/ (Arm Compiler for Embedded 6.22, /apps/arm/ArmCompilerforEmbedded6.22/bin); ELF at software/build/<test>/host0/app.v8-a.elf; flat image with fromelf --bin. Scatter files: default (code in flash at 0x40_0000, data/stack/heap in SRAM at 0x80_0000–0x80_7FFF) and dram (single load region at 0x8000_0000, stack/heap in SRAM 0x80_0000–0x80_3FFF). The on-chip RAM window the NIC-400 decodes is 0x80_0000–0x80_FFFF.
RTL simulation of the FPGA design	VCS 2022.06; scratchpad n25/sim: files.f (Vivado compile order + unisims IBUFDS/BUFG/BUFGCE + glbl), tb_haps.sv (UART0 data-register tap, bus monitors), ~45 s CPU per 400 ms of SoC time for the cache programs; dap/ and uart/ testbenches add DAP-Lite port, debug-APB, UART1 line and interrupt monitors and a 300 ms watchdog; ROM images from scratch test directories via bootrom_gen.py; interrupt-heavy tests cost ~11 s CPU per ms.

Appendix B — Build index

Builds 19–32 (29 July – 17 September) are listed in rev4 Appendix B and not repeated. Builds since (all archived with bitstream, reports and notes under FPGA_Tasos/haps_top/bitstreams/ from build 43 on):

Build	Machine	What changed	Outcome
33	Linux	QSPI generated clock at its true source + set_clock_groups -asynchronous	Routing failure (partial-route conflicts) after ~11 h
34	Linux	Same clock + set_max_delay -datapath_only 200 ns (no -from)	Built; 115,442 failing endpoints
Pre-A	Windows	Port to Vivado 2026.1 only	WHS -4.592; 9 latches
A	Windows	Gating bypass on PCK-600/GIC-400; resets; ROM stall; L2 RAM; logger; peripherals	WHS -3.524
B	Windows	QSPI reclocked; all latches removed	WNS +25.909, WHS -0.664, 0 latches, 0 critical warnings. Recommended for first hardware test
C, D	Windows	+ flash I/O constraints; post-route hold fixing; global hold iteration	WHS -0.856; hold iteration had no effect
E	Windows	Flash pad hold exclusion	Running at time of writing

36	Linux	Independent application of the 23 September proposal + QSPI clock at register output with pair-wide exceptions	WNS 0.000, WHS -3.940, 26,980 failing endpoints
36B	Linux	As 36 with the QSPI clock declared at the mux output and a pad clock	WHS -7.604 (pad hold artefact), 22,297 failing endpoints
37	Linux	Merged tree: Windows E sources + RT clock fix	WNS +13.309 (flash input path; internal +21.7), WHS -0.649, 44,955 failing hold endpoints, 0 latches, 7 clock buffers all flop- or pad-driven. Bitstream archived as haps_top_build37_merged_20260930.bit
38–41	Linux	Whole-bus logger with sticky milestones and sampled fetch address	Located the boot loop: SRAM reads return zero (bug 51)
42	Linux	Windows Build G merge (timing closed) + SRAM wrapper polarity fix	WNS +21.5, WHS +0.01. First banner on the console, 10 Oct. Stops at enable_caches()
43	Linux	Caches-off diagnostic ROM; logger bus map 43	Reaches the flash jump; synchronous exception at 0x400000
44	Linux	Diagnostic ROM v2: JEDEC, registers, exception syndrome; stuck indicators	Flash answers EF 60 16 (Winbond); controller QPI vs chip SPI; UDF at 0x400000
45	Linux	Monitor ROM + haps_qspi.py host tool	UART RX proven; sessions cut by the 10 s auto-reset
46	Linux	Auto-reset hold-off on console activity; ROM v3 (barriers)	Interactive sessions; flash writes still ignored
47	Linux	512 KB block-RAM slave on the DRAM port; dram scatter file	Load works; execution faults at +4 (first-beat bug)
48	Linux	RAM slave first-beat fix	WNS +19.447. hello_world PASS from RAM, 15 Oct; 7 of 11 tests PASS
49	Linux	In-FPGA SST26VF064B flash emulator; ROM v4 production boot	WNS +29.32. 7 of 7 from the emulated flash via unmodified SPI_STARTUP
50	Linux	Controller 1-bit write fixes; real flash pins; ROM v5 with Winbond start-up	WNS +15.067. QE set, page program OK; hello_world from the physical flash by hand, 19 Oct
51	Linux	ROM v5.2: corrected Winbond start-up (dummy 6, robust exit), auto-boot	7 of 7 from the physical flash, 20 Oct
52	Linux	UART0/UART1 cross-wiring, PL011 loopback	WNS +20.349. pl011_uart_tests PASS; uart_tests defects identified
53	Linux	L2 data RAM model 1-cycle	WNS +16.332. Cache stop unchanged
54	Linux	Arm generic L2 tag integration + ca53_generic_ram_fpga.v	WNS +14.072. Both caches work; stock boot from the physical flash, 21 Oct; 7 of 7 regression
55	Linux	Build 54 sources rebuilt from the GitHub work folder	WNS +15.130, WHS +0.009. 23 Oct: 7 tests + pl011 PASS from the flash with no SRAM preload; timer_tests FAIL (MBT-61); dap_tests abort syndrome EC 0x25, FAR 0x0130_0FD0

56	Linux	Stock Winbond bootloader ROM + megasoc_cpu_ss.sv debug-window fix (PADDRSYS offset, DBGROMADDR)	WNS +14.240, WHS +0.009. Boot never observed: ROM console at 625 kbaud (built from the upstream tree); archived for the stock-boot check
57	Linux	Monitor ROM v5.2 + the same cpu_ss fix	WNS +14.240, WHS +0.009. 27 Oct: dap_tests PASS (MBT-58), seven-test regression PASS, uart_tests v5 PASS (MBT-59): 10 of 10 from the physical flash
DDR4 ex	Linux	Xilinx DDR4 IP example design on the SODIMM, standalone (v1 LEDs, v2 VIO)	WNS +0.435 / +0.369, WHS +0.010. 28 Oct: calibrates on the board, traffic generator error-free (MGSOC-34 step 1)
58	Linux	DDR4 IP on the NIC-400 DRAM port through AXI clock and width converters; USE_DDR4; flash emulator generate-guarded	WNS +0.510, WHS +0.010. Calibrates inside the SoC; code executes from the SODIMM (29 Oct). With the caches on, dirty evictions corrupt: L2 DATA RAM example wrapper (row 79)
59	Linux	Arm generic (index-masked) L2 data RAM wrapper; duplicates disabled; RTL = build 58	WNS +0.448, WHS +0.010, 310 block RAM primitives. 30 Oct: ddr4_fpga_tests v8 PASS from DDR4 and flash with the caches on; 7 of 7 RAM-mode tests from DDR4; 10 of 10 flash images. Board baseline